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AI-Driven Adaptation for Inclusive HCI

A Dynamic Context-Aware Framework Design and Validation

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Abstract

Digital interface design has long faced an incompletely resolved challenge: how to sense a user's cognitive state in real time without specialised hardware, and respond in a manner that is transparent and reversible, while placing users with cognitive accessibility needs at the centre of the design rather than treating them as an afterthought. Existing adaptive interface frameworks exhibit notable limitations across three dimensions: real-time cognitive sensing, user control mechanisms, and inclusive coverage.

This thesis proposes the Dynamic Context-Aware Adaptive Framework (DCAF) and preliminarily validates it through a Design Science Research methodology. The framework decomposes the adaptation process into three layers with clearly bounded responsibilities: a sensing layer that continuously and passively collects user behavioural signals; an inference layer that handles both explicit preferences and implicit behavioural inference through a dual-AI architecture; and a renderer that translates declarative decisions into interface changes through a four-channel execution mechanism. The research unfolds across two iterative design cycles, running in parallel with a recruitment questionnaire study of 62 respondents. The framework is instantiated as an adaptive email client and evaluated through a within-subject controlled experiment with ten participants and nine semi-structured interviews.

Results indicate that the DCAF-powered adaptive client produced positive outcomes across both objective performance and subjective experience dimensions: task completion time decreased by a mean of 27.4%, email opening events fell by 35.8%, mouse movement complexity dropped by 19.6%, and the System Usability Scale mean reached 74.8. Effects were most pronounced among participants with attention-related difficulties. Qualitative analysis revealed that transparency and reversibility function as prerequisite conditions for user acceptance of adaptive systems, and that users' evaluations of different adaptation features follow a consistent value hierarchy.

The contributions of this research lie in providing a transferable architectural specification for adaptive interface systems, and in empirically confirming the necessity of mixed-initiative design in AI-driven interface systems.

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As this thesis comes to an end, I have slowly begun to realize that my two years in Sweden are also coming to an end. Looking back now, everything seems to have passed much faster than I once imagined. When I first arrived here, almost everything felt unfamiliar — the long winters, the quiet streets, the different rhythm of life, and facing most things alone. Over time, those ordinary and repetitive days gradually became part of my daily life without me noticing. Only now, while writing these final pages, do I realize how deeply these moments have stayed with me.

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1 Introduction

1.1 Motivation

Digital systems have long been woven into nearly every aspect of modern life communication, commerce, healthcare, and beyond. At the centre of these systems sits the user interface: both the exit point for information, presenting graphical, textual, and audio content to users, and the entry point for intent, translating human actions into machine-readable instructions [3]. How an interface is designed shapes who can access digital services, how efficiently they can use them, and whether the experience leaves them feeling in control or simply frustrated [4].

Over the past three decades, interface design has gone through several shifts as device form factors have continued to fragment. In the early days of GUI computing, design was largely device-specific—separate layouts for desktop monitors, laptops, then mobile phones and tablets, each developed independently for its screen dimensions [5]. This approach was manageable when the device landscape was limited, but as form factors multiplied, it became increasingly untenable. The emergence of responsive web design in the late 2000s was a genuine turning point: through fluid grids and media queries, interfaces could automatically adjust their layout to screen size and orientation [6]. Responsive design quickly became the norm in the industry. It cut down on development costs and made things much more consistent across devices [7].

More recently, personalisation has become a main theme in digital product design. E-commerce platforms draw on purchase histories, streaming services arrange content according to viewing preferences, and short-video apps curate feeds based on inferred interests—all leveraging user data to shape what gets delivered [8, 9]. It is worth being precise here, though: what these systems personalise is primarily content, not interface. The distinction is not merely semantic. Karat et al. [10] drew this line as early as 2003, personalization and customization are two different things: the former is when the system automatically adjusts based on inferred user characteristics, while the latter is when the user actively modifies the interface settings. By either definition, the examples above operate at the content layer. The underlying navigation structure, interaction logic, and layout remain largely identical across users. What feels like a personalised experience

is, in most cases, the same interface serving different content.

Some attempts have been made to push further. Server-Driven UI (SDUI), now widely adopted in industry, allows servers to dynamically deliver interface descriptions that clients render as distinct layouts and components, enabling meaningful interface variation across users and contexts. This represents arguably the most ambitious current approach to interface-level differentiation, and it has made dynamic interfaces technically tractable at scale. Even so, the logic driving SDUI adaptations typically comes from business rules and A/B testing outcomes—not from any sensing of a user's real-time cognitive state.

That gap is what responsive design and existing personalisation approaches both leave open. Responsive design adapts to the device, not the person; content personalisation adapts to historical behaviour patterns, not to what is happening right now [11]. Neither has the means to respond to a user's cognitive state in a given moment—the attentional drift that accompanies fatigue, distraction, or time pressure, or something more persistent like an attention regulation difficulty [12].

The idea of adaptive user interfaces (AUIs) is not new. Since the 1990s, researchers have explored systems that adjust their appearance or behaviour based on user models or contextual signals [13]. Early work depended on rule-based systems, which made changes happen automatically when certain conditions were met. For instance, new users could not see advanced features, or menus could be rearranged based on how often they were used [14]. Later, context-aware computing made it possible for systems to recognize things like location, time of day, and device type and act accordingly [15]. Smart home lighting that adjusts to the hour, or screen brightness that tracks ambient light, are familiar examples. These adaptations are useful as far as they go—but they are preset and coarse-grained, with little capacity to infer anything about a user's real-time cognitive or attentional state.

Meanwhile, research on attention-aware systems has established that detecting user engagement through behavioural signals—gaze direction, mouse movement, interaction patterns—is technically feasible [12]. Reading dwell time, switching frequency, and scrolling behaviour can all serve as indirect indicators of cognitive load or distraction [16]. In this sense, attention-aware sensing can be understood as a specialised form

of context-aware computing—one that treats the user's cognitive state as the relevant context—and adaptive user interfaces as the execution mechanism that acts on what that sensing reveals. Together, these three strands make real-time, cognitively-informed interface adaptation technically conceivable. The problem is that they have largely developed in parallel without converging into a unified, reusable framework for building accessibly adaptive systems. Existing solutions also tend to address only one side of the problem: either accessibility features for users with specific impairments, or personalisation algorithms for the general population, with relatively few attempts to tackle both at once [17]. There is a real gap between the technical capacity to sense user context and the ability to translate that sensing into inclusive, user-controlled adaptive behaviour.

This thesis is an attempt to close that gap. It proposes and preliminarily validates a candidate architecture—the Dynamic Context-Aware Adaptive Framework (DCAF)—which formalises the adaptive process as a structured pipeline: beginning with multimodal context sensing, progressing through a central inference component, and culminating in dynamic interface adaptation. The central question is whether this kind of structured approach is feasible and can guide the development of functional adaptive prototypes. The argument is worked through via a specific case study, which serves to evaluate the framework's viability, surface potential failure modes, and map out the boundaries of its applicability.

1.2 Research Questions

This research is guided by the following primary question and its subsidiary questions, which are designed to empirically evaluate the feasibility and characteristics of the proposed Dynamic Context-Aware Adaptive Framework (DCAF):

RQ1: Which aspects of the DCAF can be successfully instantiated into a functional prototype that realizes its core "sense–infer–adapt" pipeline?

Success is developing a working "focus-aware" email client prototype in which observable user interactions trigger predefined, rule-based interface adaptations.The focus is on identifying which components of the DCAF (e.g., sensing module, inference engine, adaptation logic) can be operationalized in a real system.

RQ2: What are the effects of a DCAF-based adaptive system on user performance, as measured by objective metrics (e.g., task completion time and accuracy), compared to a static baseline?

This question focuses on quantifying the impact of the adaptive system on task efficiency and correctness.

RQ3: How do users perceive and experience a DCAF-based adaptive system, as captured through subjective feedback (e.g., perceived helpfulness, controllability, and overall usability), compared to a static baseline?

This question explores the qualitative aspects of user interaction, focusing on subjective perceptions of support, control, and overall usability.

1.3 AI Usage Statement

Generative AI tools were used at several stages of this thesis project. The primary tool was Claude (Anthropic, claude-sonnet-4-6); DeepL was used solely for translation assistance. Specific uses are detailed below.

Writing-related. Claude assisted with grammar checking and Chinese-to-English translation throughout the writing process. All academic arguments, analyses, and conclusions were formulated independently by the author, who reviewed, revised, and takes full responsibility for all text presented in this thesis.

Interview design and data processing. Claude assisted with the structural planning of the interview guide; the content and direction of interview questions were determined independently by the author. Following the interviews, Claude was used to generate meeting summaries from audio transcripts as a supplementary measure to reduce the risk of omission; all qualitative analysis and interpretation were conducted independently by the author.

User study materials. Claude assisted with the layout and formatting of the task sheets used in the user study; task content, difficulty levels, and cognitive demand parameters were determined independently by the author.

Literature review. Claude was used to generate content summaries of selected refer-

ences as a supplementary aid to reduce the risk of misreading or omission; the selection, analysis, and argumentation of all literature were conducted independently by the author.

Prototype development. The email dataset used in the prototype was generated with Claude; core themes, content types, cognitive load levels, and task relevance parameters were specified in advance by the author. Claude also assisted in generating portions of the stylesheet files and in identifying and resolving code bugs during prototype development.

All AI-generated content was reviewed, revised, and approved by the author prior to use. The author bears full responsibility for the entire content of this thesis.

1.4 Delimitations

The scope of this research is defined by a number of deliberate design choices.

At the level of **technical implementation**, DCAF is instantiated in this research exclusively as a web-based desktop application. Implementation pathways for mobile, native, or cross-platform frameworks fall outside the scope of this study. The sensing layer's signal collection relies entirely on software-level behavioural events; specialised hardware inputs such as eye-tracking, physiological sensors, or cameras are not considered.

At the level of **participants background**, this research focuses on two groups: adults with attention-related difficulties or cognitive accessibility needs, and general adult users. Children, older adults, and other specific ability groups were not included in the recruitment scope; the compatibility of their needs with the framework awaits independent investigation.

At the level of **evaluation scope**, this research validates the framework within a single application scenario—an email client—with the goal of demonstrating technical feasibility and generating preliminary design insights rather than establishing statistically generalisable conclusions. The framework's performance in other application domains is not within the evaluation scope of this research.

At the level of **AI model selection**, this research calls an external LLM via the OpenRouter API and does not involve model training, fine-tuning, or local deployment. The implications of dependence on a specific model are treated as a research limitation rather

than a delimitation.

1.5 Thesis Outline

This thesis comprises six chapters, organised as follows.

Chapter 2 reviews the theoretical foundations and related literature relevant to this research, covering human cognitive load and attention theory, accessibility and inclusive design, the concepts and classification of adaptive user interfaces, and context-aware computing. Each section concludes with an analysis of its connection to the design goals of DCAF.

Chapter 3 introduces the methodological framework and overall research process. Building on an account of the Design Science Research methodology, it describes the progression of two iterative design cycles and a parallel user recruitment study, and presents the detailed design of the user study.

Chapter 4 presents the research results, covering in sequence the framework architecture and prototype implementation of the current version of DCAF, a structural limitation analysis of the first-generation framework and its role in driving the second iteration, the data analysis from the recruitment questionnaire study, and the quantitative and qualitative results of the user study.

Chapter 5 discusses the research findings across four dimensions: answers to the research questions, design implications, research limitations, and ethical considerations, and proposes directions for future work.

Chapter 6 concludes the thesis, restating the research contributions and situating this work within the broader research landscape of adaptive user interfaces.

2 Theory and Related Work

2.1 Theory

2.1.1 Human Cognitive Load and Attention

The Nature of Attention At its core, human perception and cognition operate under hard resource constraints—and attention is the mechanism that makes functioning under those constraints possible. This process allows us to focus and interact with the most important information from the constant stream of environmental data, while pushing less important details to the background of our awareness [12]. Without this filtering capacity, the cognitive system would simply be overwhelmed. As Chun and Wolfe [18] put it plainly: ”at any given time, the environment presents far more perceptual information than can be effectively processed” (p. 273).

Cognitive psychology researchers have used different metaphors to explain how attention works, each highlighting a different aspect of its function. The "spotlight" metaphor is probably the most well-known; it suggests that attention works like a beam of light, and things within this beam are processed more thoroughly than those outside it [19]. The ”zoom lens” model extends this idea further: the beam’s diameter is adjustable, narrowing to a small area for intensive, concentrated processing, or widening to cover a broader field at lower intensity [20, 21]. Both are simplifications, of course, but they capture something real: attention is not evenly distributed.

A further distinction worth drawing is between the two modes in which attention can be driven. Researchers generally distinguish between endogenous attention (top-down, goal-directed) and exogenous attention (bottom-up, stimulus-driven) [12]. The former is what we choose to do on purpose, like focusing on a certain part of a text or scanning a page for a certain item. It takes work and long-term involvement. Exogenous attention, by contrast, does not wait for a decision: a sudden animation, a brightly coloured button, or an unexpected notification sound can redirect our gaze before we have consciously registered it. As Chun and Wolfe [18] describe it, endogenous attention ”is voluntary, effortful, and has a slow (sustained) time course,” while exogenous attention ”draws attention automatically and has a rapid, transient time course” (p. 279).

In practice, these two modes are always interacting. Our goals determine what we are looking for; prominent stimuli in the environment can interrupt that search and pull attention elsewhere. For interface design, this dynamic has fairly direct implications: designers need to simultaneously support users' goal-directed focus and manage the potential disruptions generated by the interface itself.

Measuring Attention For adaptive systems that aim to respond to a user's cognitive state, the question of how attention is measured becomes practically significant. If behavioural signals such as gaze direction or reading dwell time are to serve as proxies for attentional focus, it is necessary to understand both what these signals can and cannot reliably indicate.

Eye-tracking is the most common measurement approach, using gaze direction as a proxy for attentional focus—but looking at something is not the same as processing it [21]. Inattentional blindness illustrates this: when attention is directed elsewhere, a fully visible object can fail entirely to enter conscious awareness [22, 23]. As Wood et al. [21] note, gaze direction alone is insufficient to accurately indicate what information has actually reached the user's awareness (p. 592).

Related to this is change blindness—the tendency for visual changes occurring during eye movements or brief interruptions to go completely unnoticed [24]. Together, these phenomena suggest that visual awareness is considerably narrower than subjective experience implies, and that attentional state cannot be read off directly from where a user appears to be looking.

Triesch et al. [25] provide a further illustration: participants performing a block-sorting task frequently failed to notice that blocks had changed colour between selection and placement, because colour was task-relevant only during selection. Attention tracks task demands rather than the perceptual field as a whole. For DCAF, this has a direct design implication: behavioural sensing must be interpreted in relation to task context, and the system cannot assume that information placed on screen has been perceived by the user.

Cognitive Load Theory Attention research is concerned with how information gets selected; cognitive load theory deals with what happens next—specifically, how much

working memory can actually handle once information has been taken in. Sweller [26] developed this framework on the premise that working memory is severely limited in capacity, while long-term memory is effectively unlimited.

The theory identifies three distinct types of cognitive load that combine to determine the total pressure on working memory at any given moment [27].

Intrinsic cognitive load comes from the inherent complexity of the material being processed—more precisely, the number of elements that must be held in mind simultaneously and the degree to which they interact [27]. Learning vocabulary carries relatively low intrinsic load, since each item can be handled independently; understanding a complex electronic system is another matter entirely, requiring the simultaneous maintenance of many interdependent concepts. Importantly, intrinsic load is not fixed—it shifts with the learner's level of expertise. Content that is cognitively demanding for a novice may be effortless for an expert who has already developed schemas that chunk multiple elements into single manageable units [28].

Extraneous cognitive load arises from how information is presented, and from the ancillary operations a task requires that are not directly relevant to learning or task completion [29]. A confusing navigation structure, or an interface that requires users to mentally integrate information scattered across multiple locations, imposes cognitive costs that have nothing to do with the task itself. Hollender et al. [29] further distinguish, in the HCI context, between load generated by instructional design and load generated by the software itself—a distinction that is particularly relevant for adaptive systems, since the interface can either reduce or amplify cognitive burden depending on how it is designed.

Germane cognitive load refers to the cognitive resources invested in schema construction—the mental work of actually understanding and internalising new information [27]. Unlike extraneous load, germane load is beneficial: it reflects the mind doing useful work. As Sepp et al. [28] put it, germane cognitive load "arises from active schema construction processes, and is therefore beneficial for learning" (p. 296).

The three types are additive. Intrinsic load sets the baseline complexity; extraneous load consumes resources that could otherwise support understanding; germane load represents the productive use of whatever capacity remains [30]. For the system to

function effectively, total load must stay within working memory's limits.

2.1.2 Accessibility and Inclusive Design

A Landscape of Overlapping Concepts Accessibility and inclusive design are not a single unified field so much as a cluster of related but distinct traditions, each with its own historical roots and practical emphases. Kille-Speckter and Nickpour [31] provide a systematic timeline of this evolution, tracing the narratives and milestones from early design for disability to contemporary inclusive design. Persson et al. [32] further observe that terms such as "universal design," "inclusive design," "accessible design," and "design for all" share broadly similar ambitions but differ in both method and underlying philosophy.

Accessible design has its historical roots in the disability rights movement and associated legislation, with the Americans with Disabilities Act serving as a key policy milestone [33]. ISO/IEC Guide 71 [34] frames it as extending standard design to accommodate people with functional limitations—through products that work without modification for most users, through adaptable configurations, or through standardised interfaces compatible with assistive technologies. The approach acknowledges human diversity, but in practice it has sometimes treated accessibility as a supplement to a "normal" baseline rather than as a starting condition.

Universal design was introduced by Ron Mace in 1996 and elaborated in his work [35]. Its definition is direct: design products and environments so that as many people as possible can use them without adjustment or specialisation. The North Carolina State University Center for Universal Design elaborated this into seven operating principles [36]. Mace's own observation cuts to the point: what removes a barrier for one person may create one for another, and simply eliminating obstacles is not enough [35].

Inclusive design, more common in British contexts, is formalised in BS 7000-6 [37]: designing mainstream products and services to be accessible to as wide a range of people as possible, without special adaptation, as is reasonably possible. Design for All, prevalent across continental Europe, is framed by the EIDD Stockholm Declaration [38] around equality and participation: everything designed and made for human use must be accessible and responsive to evolving human diversity, with end users actively involved

at every stage of the design process.

The conceptual differences between these terms are directly relevant to this paper. One reason for the ambiguity is that people tend to view accessibility as an isolated attribute of the user or the product itself, rather than a combined result of the relationship between the two. This paper is designed to address this issue to some extent: instead of categorizing users into fixed accessibility classes, the framework aims to dynamically respond based on the user's actual capability state at a given moment—serving both users with cognitive accessibility needs and ordinary users in an adaptive architecture.

The Challenge of Defining and Measuring Accessibility At the centre of these debates is a question with direct practical significance for this thesis: what does it mean for something to be accessible, and how would one measure it? Without a workable definition and means of measurement, it is difficult to evaluate whether an adaptive system genuinely serves the users it targets.

ISO 26800 [39] defines accessibility as the degree to which a product, system, service, environment, or facility can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use. This connects directly to ISO 9241-11's [40] definition of usability. As Persson et al. [32] draw the distinction: usability is about the quality of the experience for a particular user group; accessibility is about how wide that group can be.

What this framing suggests is that accessibility is not a binary property. It is dynamic and context-dependent. Persson et al. [32] argue that accessibility should not be a question of "what am I," but rather "what role do my varying and changing capabilities play in this particular situation." Measuring that gap requires integrating technical, medical, social, and humanistic knowledge simultaneously [32]—which is part of what makes it difficult, and part of what makes the dynamic, context-sensitive approach that DCAF takes a meaningful contribution to this challenge.

Digital Design Marginalization Sin et al. [41] introduce a concept that sharpens this analysis considerably: digital design marginalization (DDM). Their definition is worth quoting closely—DDM is "the process by which digital interface design excludes

certain users and leads to their marginalization in other areas of their lives" [41]. The key move here is from passive exclusion to active pushing away. As they put it, "pushing away" better captures a non-binary reality than "excluding" does: some users may manage to overcome the marginalizing effects of a design without being fully excluded, but they are still pushed—and the experience of being pushed still carries consequences [41].

Dwivedi et al. [42] originally defined marginalization in terms of limiting access to the benefits of globalization, whether intentionally or not. Sin et al. [41] reframe this for the design context: DDM describes design decisions whose consequences extend beyond a user being excluded from a specific system, producing additional negative social outcomes—greater difficulty using non-digital equivalents, reduced capacity to interact with other digital systems, or increased barriers to previously accessible services, whether social, financial, or administrative.

Their case studies make this concrete. An app designed for older adults with the explicit goal of enhancing social connection—InTouch—managed to produce something close to the opposite: family tension, reduced self-efficacy, and increased risk of social isolation. The reasons are instructive: the design failed to account for all the social actors involved, misread users' mental models, ignored hardware accessibility beyond the software itself, and did not anticipate the downstream effects of its design choices.

One finding is particularly worth highlighting. Franz et al. [43] found that some users actively avoided accessibility features because they associated them with disability—a label they did not apply to themselves. One participant expressed the hope never to need such features, precisely because of the stigma attached to them. This creates a situation with no good exits: the user can either struggle with an inaccessible mainstream interface, risking exclusion from the services it provides, or use the accessibility features and experience the social cost of doing so. Both paths, in other words, lead toward marginalization [41]. It is a reminder that the problem of inclusive design is not simply technical—the framing and labelling of features carries its own weight.

2.2 Related Work

2.2.1 Adaptive User Interfaces: Concepts and Classification

Core Concepts and Research Orientations The concept of adaptive user interfaces (AUIs) originated in the early 1990s. Schilit and Theimer [44] initially introduced the term "context-awareness" in mobile computing, providing a theoretical foundation for the discipline. Miraz et al. [45] define an adaptive interface as ”a software artefact that dynamically changes its characteristics and functionalities based on the user’s current context, in order to enhance interaction effectiveness.” Their survey organises the research landscape into three broad orientations: AI-focused work, which develops reasoning mechanisms that monitor user behaviour to identify patterns and provide intelligent assistance; user modelling work, which constructs representations of user characteristics and preferences as the basis for adaptation; and HCI-focused work, which attends to multimodal interaction and the communication channels between system and user. For classification, Langley [46] offers a framework based on the type of user feedback involved: *informative* interfaces filter and modify information, presenting only what is relevant to the current task; *generative* interfaces focus on producing useful knowledge structures, such as document preparation or planning tools.

Adaptive vs. Adaptable: A Distinction Worth Drawing Browne [13] makes a clear distinction between "adaptable" and "adaptive" interfaces. Adaptable systems give users control and let them change the settings based on their own preferences. Adaptive systems, on the other hand, take the lead and figure out what the user needs without any input from them. Each approach carries its own costs—system-driven adaptation reduces user burden but sacrifices transparency; user-driven customisation preserves a sense of control but demands additional cognitive effort. Stephanidis et al. [47] proposed a ”mixed-initiative” model that attempts to occupy the middle ground, where both user and system participate in the adaptation process, with the user free to intervene when the system acts, and free to initiate changes independently.

Rule-Based Adaptive Systems One of the first widely used methods was rule-based ones. The logic is simple: set up a set of if-then rules ahead of time and make the

necessary interface changes when the conditions are met. The AVANTI project [47] is an early and well-documented example—when the system inferred limited short-term memory, it responded by simplifying language, reducing animation, and condensing information blocks. The primary strength of this approach is transparency: the logic is explicit and debuggable [48]. The limitation is equally clear—rules must be written in advance, and they tend to struggle with situations that were not anticipated.

Machine Learning and User Modelling Approaches As machine learning matured, data-driven methods began to occupy a growing share of the adaptive systems landscape. Nivethika et al. [49] proposed a K-means clustering framework that groups users by interaction patterns into experience levels—novice, intermediate, expert—and renders corresponding interface variants, requiring no labelled data. The Lumiere project by Horvitz et al. [50] used Bayesian networks to reason about the inherent uncertainty in inferring user needs, drawing on background characteristics, operation history, and queries to determine what kind of assistance to offer—work that later fed into the Microsoft Office Assistant. Compared to rule-based systems, probabilistic methods handle ambiguous signals more gracefully, but tend to do so at the cost of interpretability.

Overall, these works trace the evolution of adaptive interface research and reveal a limitation that remains largely unresolved: existing systems tend to excel along one dimension—transparent but rigid, flexible but opaque, system-driven or user-controlled—without bringing these capabilities together in a unified architecture. DCAF is, in part, a response to this gap. It instantiates the mixed-initiative principle [47] as a dual-AI architecture, with the Settings AI handling explicit user preferences and the Base AI processing implicit behavioural signals; it retains a rule-based inference engine as a core periodic path, directly reflecting the user preference for predictability and transparency documented in Chapter 5. At a more foundational level, the work of Nivethika et al. and Horvitz et al. supports a premise that DCAF depends on: that meaningful user state inference is achievable from interaction data alone, without requiring explicit self-report from the user at every step.

2.2.2 Context-Aware Computing

Defining Context The intellectual roots of context-aware computing reach back to Mark Weiser's [51] vision of ubiquitous computing: "the most profound technologies are those that disappear—they weave themselves into the fabric of everyday life until they are indistinguishable from it" (cited in Baldauf et al., [15]). The systems this vision points toward are ones that sense environmental change and adjust behaviour automatically, without requiring explicit user intervention [15].

"Context" itself has been defined in many ways, each with a different emphasis. Schilit and Theimer [44] framed it as location, the identity of nearby people and objects, and changes to those objects. Ryan et al. [52] narrowed it to location, environment, identity, and time. Dey [53] took a broader view, encompassing emotional state, attentional focus, and the people and objects in the user's surroundings. Brown [54] offered perhaps the most pragmatic definition: context is simply what the computer knows about the user's environment (cited in Baldauf et al., [15]). The divergence across these definitions is itself informative—the boundaries of "context" are not fixed, and what a system treats as contextual depends on what it is trying to sense.

On classification, Baldauf et al. [15] distinguish between external (physical) dimensions—location, light, temperature, measurable directly by hardware sensors—and internal (logical) dimensions, captured primarily by monitoring user interaction, such as task state, goals, and emotional condition. Hoareau and Satoh [55] further note that context can include non-sensed information, such as personal memory or current emotional state—information that cannot be directly measured but is often essential for understanding what a user actually needs. They characterise context as rich, subjective, and constantly changing: any system that attempts to capture it must contend with all three.

System Architectures Context-aware systems can be implemented in several ways. Baldauf et al. [15] identify three main approaches to context data acquisition: direct sensor access, where client software reads from hardware sensors directly—tightly coupled and suitable for limited scenarios; middleware infrastructure, which introduces a layered architecture that separates application logic from low-level sensing details, improving extensibility; and context servers, which move data collection to a remote component that

multiple clients can access concurrently, reducing the burden on resource-constrained devices.

At the architectural level, Baldauf et al. [15] propose a five-layer conceptual model: a sensor layer for data acquisition; a raw data retrieval layer that accesses sensors via drivers and APIs; a preprocessing layer that interprets and abstracts technical data—combining temperature and light readings to infer, for instance, whether a user is indoors or outdoors; a storage and management layer that organises data and exposes it through a common interface; and an application layer that implements context-responsive behaviour. The organising principle is separation of concerns: each layer handles only what it needs to, and upper layers remain insulated from the implementation details of the layers below.

Context Modelling How context information is represented is a design question in its own right. Hoareau and Satoh [55] survey six main approaches. Key-value models are the simplest—an identifier maps to an attribute value—with limited expressiveness but good scalability. Markup scheme models use hierarchical tag structures; CC/PP and UAProf are standard examples, well suited to describing device capabilities and user profiles. Object-oriented models apply encapsulation and inheritance to represent context entities and their relationships. Logic models use facts and rules to define context, supporting first-order reasoning [56, 57]. Ontology-based models are currently the most expressive: Strang and Linnhoff-Popien [58] evaluated six modelling approaches against a set of requirements and concluded that ontologies ”have the most expressive model, satisfying most of the requirements.” Chen et al.’s [59] CoBrA-ONT ontology, implemented in OWL, provides both a vocabulary for context description and a basis for cross-system data sharing. Loke [60] further developed the logic programming approach, exploring the integration of rule-based context behaviour triggers with context infrastructure.

The context-aware computing literature provides important background for this thesis, though the research direction taken here differs from the classical focus of the field. Most context-aware systems are concerned with physical environment parameters—location, time of day, device type. This thesis is instead concerned with the user’s cognitive state, which corresponds to what Baldauf et al. [15] call the internal (logical) dimension of context, and to the attentional focus and emotional state that Dey [53] includes in his broader definition. Applying the methodology of context-aware computing to the user’s

internal state rather than their physical environment is one of the core design orientations of the framework proposed here.

2.2.3 Existing Frameworks and Their Limitations

Placing these three frameworks side by side, several shared limitations become clearer.

On real-time cognitive sensing: Peissner et al.'s [61] adaptation logic is grounded in static user profiles and cannot respond to moment-to-moment state changes; Carrera-Rivera et al.'s [62] contextual signals—user role, time of day—are too coarse-grained to capture cognitive load or attentional state; Machado et al. [63] achieve real-time monitoring but only through specialised eye-tracking hardware that most consumer deployments cannot support.

On transparency and user control: MyUI's animated transitions and confirmation dialogs did not, in practice, resolve the comprehensibility problem for its target users [61]; AdaptUI's neutral control scores suggest users largely accept recommendations rather than directing them [62].

On user coverage: none of the three frameworks specifically address users with cognitive accessibility needs—those who may experience attention regulation difficulties or elevated cognitive load in everyday digital interactions.

Table 1 summarises these dimensions across the three frameworks.

Framework	Hardware	Sensing	Transparency	User Coverage
Machado et al. (2018)	Eye-tracker	Physiological signals	Not addressed	Older adults
Peissner et al. (2012)	None	Static profile data	Animated transitions	Multi-group users
Carrera-Rivera et al. (2024)	None	Interaction + context signals	Recommendation explanation	General users

Table 1 Comparison of adaptive UI frameworks across key dimensions

These shared limitations point toward a design space that existing frameworks have not fully occupied: a system capable of real-time cognitive sensing without specialised hardware, offering transparent and reversible adaptation, and treating users with cognitive accessibility needs as a core design consideration rather than an afterthought. How to satisfy all three conditions within a single coherent architecture is the motivating

challenge that Chapter 3 addresses.

3 Methodology

3.1 Research Design

Design Science Research (DSR) is a research paradigm systematised by Hevner et al. [64] as a core methodology in information systems and human–computer interaction. Its fundamental premise is that solving complex HCI problems requires not only observation and description, but the generation of knowledge *through* design and construction. Unlike controlled experiments in natural science or survey-based inquiry in social science, DSR is artefact-centred: the researcher designs, builds, and evaluates a concrete artefact, simultaneously addressing a real-world problem and extracting transferable design knowledge from the process.

Hevner et al. organise DSR activity along two dimensions: relevance and rigour. Relevance requires that the artefact address a genuine problem in an applicable environment; rigour requires that design decisions be grounded in prior theory and that evaluation be conducted through systematic methods. Together, these two dimensions guard against DSR collapsing into either pure engineering development or theory disconnected from practice.

The research process in DSR typically unfolds as iterative cycles. Each cycle comprises three basic steps: identifying the problem and specifying design objectives; constructing an artefact that satisfies those objectives; and evaluating the artefact under conditions that approximate realistic use, feeding the results back into the problem identification stage of the next cycle. This build–evaluate–rebuild rhythm allows a framework and its prototype to converge incrementally under real feedback, rather than being fixed in advance under closed assumptions.

In this thesis, the research artefact exists at two levels: the DCAF framework as an abstract architectural specification, and the adaptive email client as its concrete instantiation. The former constitutes transferable design knowledge; the latter is the vehicle through which that knowledge is validated. The research process unfolds across two iterative cycles, beginning with a systematic literature review and proceeding through prototype construction, failure identification, and user needs synthesis, arriving at a current version that has been subject to structured empirical evaluation. How this process unfolds in

practice is described in the sections that follow.

3.2 Research Process Overview

The research process unfolds across two iterative design cycles, running in parallel with a user recruitment study. Figure 1 presents the complete research workflow.

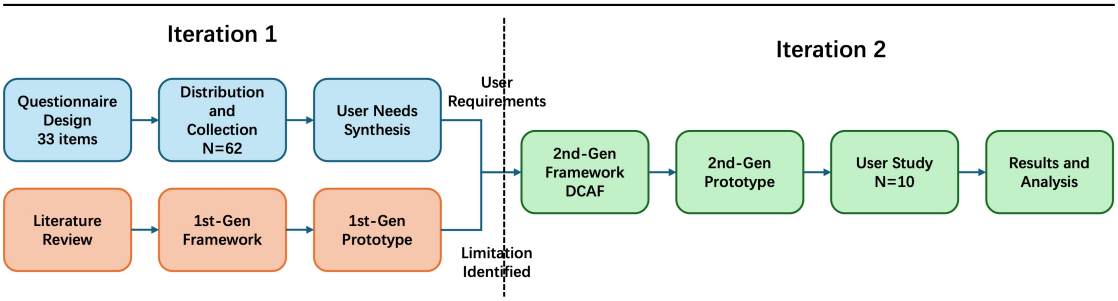


Figure 1 Overview of the research process, comprising two iterative design cycles and a parallel recruitment study.

The first cycle begins with a systematic review of three converging research streams: adaptive user interfaces, context-aware computing, and accessibility and inclusive design. This review identifies shared limitations across existing frameworks, defines the design space, and provides the theoretical grounding for the first-generation framework. The first-generation framework adopts a three-layer structure in which the inference layer submits all behavioural signals collected by the sensing layer to an AI engine, which produces a three-level cognitive load classification; the renderer then switches the interface to a corresponding preset state. A prototype is built on this basis.

Running concurrently with the first cycle, a user recruitment questionnaire study is conducted. Distributed to a pool of potential participants, the questionnaire systematically collects prior attitudes toward adaptive interfaces, acceptance of AI-driven adjustment, and privacy sensitivity toward different categories of behavioural data collection. Its findings, together with the structural limitations surfaced by the first-generation prototype, jointly constitute the design inputs for the second cycle.

The second cycle makes targeted revisions to the inference layer and the renderer, producing the current version of DCAF described in this thesis, and a second-generation prototype is developed accordingly. A formal user study is then conducted using this prototype

as its vehicle, combining objective performance measurement with semi-structured interviews.

The specific content of each cycle, the technical architecture of the first-generation prototype, and a detailed account of its structural limitations are presented in Result Chapter.

3.3 First Iteration: Literature-Driven Framework Design and Prototype Construction

The first iteration begins with a systematic review of three converging research streams: adaptive user interfaces, context-aware computing, and accessibility and inclusive design. A cross-framework comparison of existing work surfaces three shared limitations. First, existing systems rely predominantly on static user profiles or coarse-grained contextual signals, with little capacity to sense a user's real-time cognitive state. Second, the adaptation process is typically opaque to the user, lacking both interpretability and any means of reversal. Third, existing frameworks tend to address either general users or users with specific impairments in isolation, leaving a gap in coverage for users with cognitive accessibility needs in everyday digital interactions. These three limitations jointly define the design space this research addresses.

Drawing on this analysis, the first-generation framework decomposes the adaptation process into a three-layer structure. The sensing layer continuously and passively collects user interaction behaviour—mouse trajectory, reading speed, interface switching frequency, and hover events. The inference layer submits these signals to an AI engine, which produces a judgement of the user's current cognitive load state and maps it to one of three discrete levels. The renderer then transitions the interface to a corresponding preset state based on this classification. A first-generation prototype is built on this basis, serving as an initial proof-of-concept vehicle.

In practice, the first-generation prototype surfaced a number of structural limitations, spanning the granularity of the inference mechanism, the absence of an explicit user control channel, and the interaction design of adaptation feedback. These limitations are examined in detail in Chapter 4, where they also serve as one of the two primary design inputs driving the second iteration.

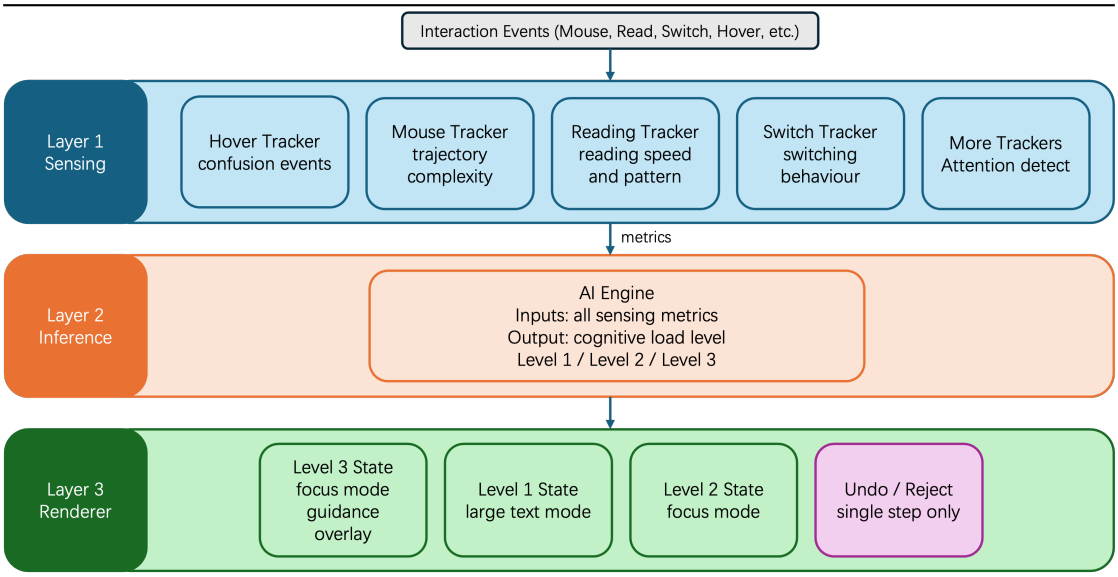


Figure 2 Architecture of the first-generation framework.

3.4 Parallel User Recruitment Study

Running concurrently with the first iteration, an online recruitment questionnaire was designed and distributed, yielding 62 complete responses. The questionnaire served a dual purpose: screening suitable participants for the subsequent controlled user study, and systematically collecting prior attitudes toward adaptive interfaces, acceptance of AI-driven adjustment, and privacy sensitivity toward different categories of behavioural data collection, providing empirical grounding for the design decisions of the second iteration.

The questionnaire comprised 33 items across five dimensions: demographics and technology background, email usage behaviour, attention and cognitive background, AI attitudes and awareness, and privacy attitudes and participation willingness. It was distributed through an online platform across multiple channels, with no mandatory screening condition requiring a formal attention-related diagnosis, in order to preserve ecological validity.

While the recruitment data do not form part of the formal experimental analysis, the attitudinal and preference findings drawn from the survey—particularly participants' expressed preferences regarding adaptation transparency, controllability, and trust in

AI-driven decisions—were used as user needs evidence to inform the design revisions undertaken in the second iteration. In this sense, the questionnaire functioned not only as a recruitment instrument but also as a lightweight needs elicitation study running in parallel with the first-generation prototype development.

The full questionnaire is provided in Appendix A. Detailed analysis of the questionnaire data is presented in Chapter 4.

3.5 Second Iteration: DCAF Framework and Prototype Refinement

The second iteration is driven by two converging inputs. On one side, the structural limitations surfaced by the first-generation prototype—the coarse granularity of the inference mechanism, the absence of an explicit user control channel, the interruptive nature of adaptation feedback, and the incompleteness of the undo mechanism—pointed toward the need for systematic revision of the inference layer and the renderer. On the other side, the user needs evidence drawn from the recruitment questionnaire study—in particular participants' expressed preferences for transparency, reversibility, and a mixed-initiative adaptation architecture—provided empirical grounding for the direction of those revisions.

The changes in this iteration are concentrated in the inference layer and the renderer. The inference layer evolves from a single-path AI engine to a dual-AI architecture. Settings AI handles preferences explicitly expressed by the user through natural language, translating them into a structured rules object. Base AI performs implicit inference over the behavioural signals collected by the sensing layer, producing a declarative adaptation decision constrained by the rules object. The two paths operate in coordination through a priority arbitration mechanism, ensuring that explicit user preferences always take precedence over behavioural inference. The renderer evolves from preset state switching to a four-channel execution architecture, supporting fine-grained CSS variable adjustment, declarative rendering of dynamic components, interface element visibility control, and progressive activation of focus mode. A complete adaptation history mechanism is introduced, supporting unlimited undo and redo across the full session; each adaptation decision is surfaced to the user through a non-interruptive bottom notification bar that

presents the reason for the change alongside a 20-second auto-accept timeout.

The resulting framework is the current version of DCAF presented in this thesis. A second-generation prototype is built on this basis and serves as the experimental vehicle for the subsequent user study. The detailed design specification of each framework layer and the full technical implementation of the prototype are presented in Chapter 4.

The second-generation prototype is built using native web technologies—HTML5, CSS3, and JavaScript ES6—running as a single-page application in a modern browser with no additional installation required. The choice of a native web stack reflects two considerations: avoiding the abstraction overhead that framework layers would introduce over the clean three-layer mapping of DCAF, and ensuring reproducibility across operating systems and devices, reducing the configuration burden of the experimental environment. The inference layer calls a large language model via the OpenRouter API, with a keyword-matching fallback engine that maintains basic adaptive functionality when the API is unavailable or the network is constrained. The two experimental versions—the static baseline and the dual-AI adaptive version—share identical visual design, email dataset, and task structure; the only difference is whether the dual-AI pipeline is active, a deliberate choice to ensure comparability across conditions.

The choice of an email client as the prototype scenario is also considered. Email processing is one of the most pervasive information-handling tasks in modern knowledge work [65], encompassing cognitive activities ranging from information retrieval and content judgement to priority decision-making. It naturally activates observable behavioural patterns and provides the sensing layer with a rich source of signals, making it well suited to the design goals of DCAF.

3.6 User Study

3.6.1 User Study Design

The user study employs a within-subject counterbalanced design, with each participant completing email tasks under two conditions in sequence: a static baseline version and the DCAF-powered dual-AI adaptive version. Condition order is alternated across participants to offset learning and fatigue effects.

3.6.2 Experimental Variables

The independent variable is interface version: adaptive versus static baseline.

Dependent variables fall into two categories.

Objective performance metrics comprise four measures. Task completion time (seconds) directly reflects the impact of adaptive intervention on overall efficiency, and is the most straightforward quantitative indicator of the system's practical value. Email opening events counts the total number of times any email is opened during a session, with each reopening of the same message counted as a separate event; this metric captures repetitive navigation behaviour under working memory pressure—users reopening the same message because they cannot recall its content, a pattern particularly pronounced among participants with attention-related difficulties—and serves as a sensitive indicator of whether the AI summary feature effectively reduces information retrieval burden. Mouse movement complexity, measured as the ratio of directional changes to total cursor distance, quantifies the purposefulness of navigation behaviour; higher values indicate more circuitous cursor paths, associated with elevated cognitive load and decision difficulty [16]. Average reading speed (words per minute) reflects the depth at which users process email content; anomalous variation in reading speed—whether unusually fast or slow—can signal information processing difficulty, and also provides a direct indicator of whether AI summaries alter users' reading patterns.

Subjective experience metrics comprise two measures. The System Usability Scale (SUS) [1] comprises ten five-point Likert items covering ease of use, learnability, and willingness to use, with 68 widely accepted as the threshold for acceptable usability. The NASA Task Load Index (NASA-TLX) [2] captures subjective cognitive load across five dimensions—mental demand, temporal demand, performance satisfaction, effort, and frustration—providing a perceived load profile that complements the objective behavioural indicators.

3.6.3 Participants and Recruitment

Ten participants were selected from the 62 recruitment questionnaire respondents on the basis of three criteria: diversity of attention profiles, range of technical familiarity, and

acceptance of behavioural data collection. The final sample is divided into two groups: the attention-related group (AR; $n = 6$; P1, P3, P5, P6, P7, P8), comprising participants with a formal diagnosis or self-reported attention-related difficulties; and the general group ($n = 4$; P2, P4, P9, P10), comprising participants with no such diagnosis or reported difficulties. Detailed participant characteristics are presented in Chapter 4.

3.6.4 Experimental Procedure

Each experimental session lasted approximately 60 to 90 minutes and was conducted either in person or via online meeting. Each session began with an informed consent procedure, during which the researcher explained that the system would collect only non-content behavioural signals—specifically mouse movement complexity and reading speed—solely for the purpose of immediate interface adaptation, with no data stored persistently beyond the session.

Each participant completed two task sets, one under each condition. Each task set comprised five structured email tasks of increasing cognitive demand, covering information retrieval, priority ranking, email composition, deadline management, and decision making. The two task sets were thematically distinct but matched in structure and approximate difficulty, to prevent participants from carrying over strategies learned in one condition directly into the next. Participants completed the SUS and NASA-TLX questionnaires immediately after each condition.

Prior to the formal study, a candidate participant not involved in the main research completed a full pilot run of the procedure. The pilot surfaced two issues requiring correction. Two questionnaire items were found to overlap in wording and were consolidated into a single more precise item. More significantly, the BaseAI inference cycle was found to be too long to trigger at all during shorter task segments; an immediate inference trigger based on reading completion events was therefore added to the system, ensuring that key interaction moments could prompt an adaptation assessment without relying solely on the periodic cycle. A further issue emerged when the pilot participant expressed discomfort at the description that the system was observing their behaviour; the informed consent procedure for the main study was revised accordingly to include a detailed explanation of the sensing mechanism, explicitly stating that data is used solely for immediate

interface adaptation and is not stored persistently beyond the session. The complete task sheets used in the study are provided in Appendix C.

3.6.5 Data Collection

Data were drawn from four complementary sources. The ResearchLogger module embedded in the prototype automatically exported a JSON session log for each condition, recording per-task timestamps, mouse movement complexity scores, and email opening event counts. The NASA-TLX questionnaire was administered immediately after each condition. The SUS questionnaire was administered after the dual-AI condition only. Semi-structured interviews were conducted after the session with nine participants (P10 was unable to participate due to time constraints), each lasting 35 to 40 minutes and covering five themes: personal experience of attention-related difficulties, everyday interface pain points, views on human–AI collaboration, reactions to the prototype, and attitudes toward AI as an embedded interface assistant. Interviews were audio-recorded with consent and transcribed verbatim. The complete interview guide and transcripts are provided in Appendix B.

3.6.6 Data Analysis

Quantitative analysis uses descriptive statistics to compare means, standard deviations, and individual variation across the two conditions. Given the small sample size, no inferential statistical tests are performed; the focus is on identifying directional trends and effect magnitudes. Qualitative analysis follows the six-step thematic analysis procedure proposed by Braun and Clarke [66], coding interview transcripts to extract recurring themes related to perceived helpfulness, controllability, comprehensibility, and inclusive experience.

3.6.7 Ethical Considerations

The study is designed in accordance with institutional ethical guidelines. All participants provided informed consent prior to participation, with clear disclosure of the research purpose, data collection scope, anonymisation procedures, and their right to withdraw at

any time. Behavioural data were collected only during the experimental session, stored locally, and contain no personally identifiable information.

4 Results

4.1 DCAF Framework and Prototype

This chapter presents the research results in four parts. It opens with a detailed account of the final DCAF framework architecture and its prototype implementation, establishing the technical foundation against which the findings are interpreted. The narrative then returns to the beginning of the design process, examining the structural limitations of the first-generation framework and how they drove the second iteration. This is followed by an analysis of the parallel recruitment questionnaire study, which provided the user needs evidence informing that iteration. The chapter concludes with the results of the formal user study conducted on the second-generation prototype, comprising both quantitative performance measurements and qualitative interview findings.

4.1.1 Framework Architecture

DCAF, as shown in the Figure 3, decomposes the adaptation process into three layers with clearly bounded responsibilities, communicating through standardised data interfaces rather than direct calls into each other's internal logic. This structural decoupling means that the concrete implementation of any layer can evolve independently without disturbing the overall framework logic.

The **sensing layer** is responsible for continuously and passively collecting user interaction behaviour, converting raw interface events into structured context signals for the inference layer. In this prototype, the sensing layer comprises four specialised trackers. The mouse tracker samples cursor position at 200 ms intervals, computing the ratio of directional changes to total movement distance as a trajectory complexity score—a proxy for cognitive load, as circuitous and uneven cursor paths are associated with decision difficulty and information overload. The reading tracker estimates reading speed and depth by monitoring how long text nodes within the viewport remain visible; anomalous reading speed in either direction serves as a potential indicator of content processing difficulty. The switch tracker records the frequency and interval of transitions between email items; high-frequency switching is a behavioural marker of decision difficulty or

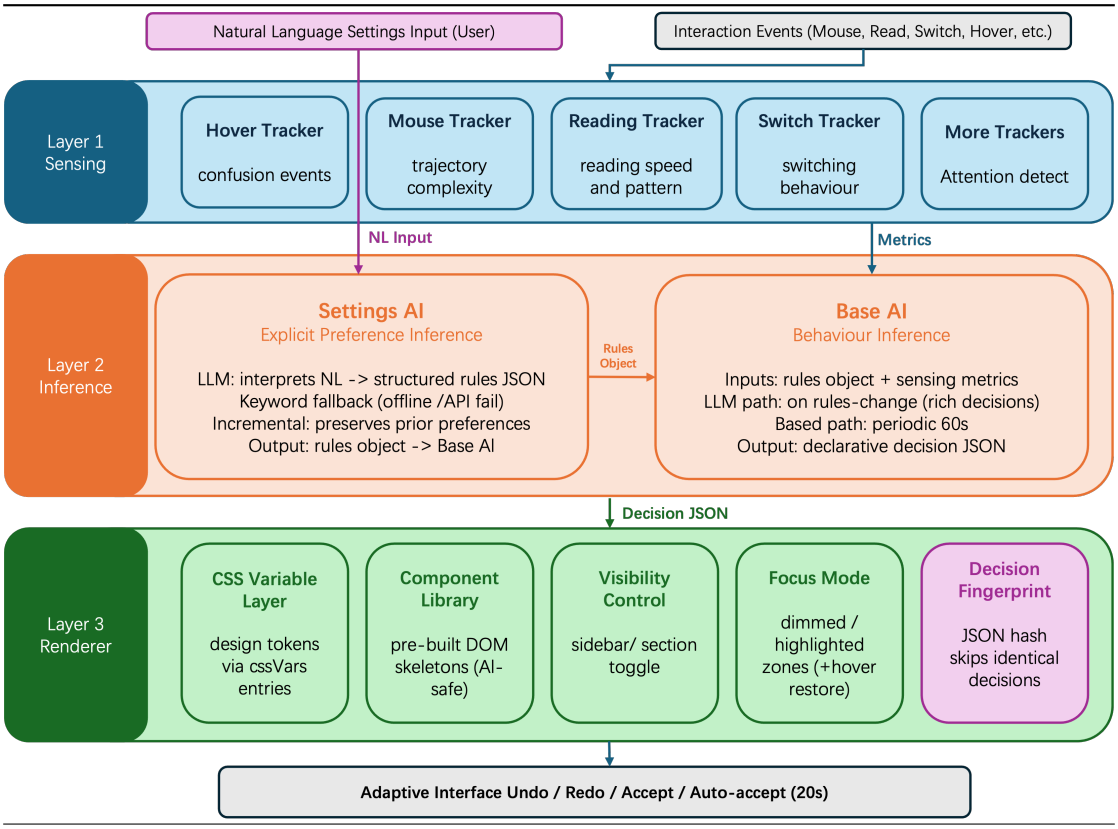


Figure 3 The three-layer DCAF architecture, showing the dual-AI inference layer and four-channel renderer.

disorganised information processing. The hover tracker captures repeated hover events over functional interface controls, interpreted as signals of preference exploration. Every 10 seconds, each tracker's aggregation method compresses behavioural events within the sliding window into a set of numerical metrics, which are written to a shared context buffer and trigger the inference layer's periodic evaluation.

The **inference layer** is the cognitive core of DCAF, employing a dual-AI architecture to handle two qualitatively different types of user signal. Settings AI handles preferences explicitly expressed by the user through natural language: the user describes their needs through a conversational input field embedded in the interface, and Settings AI calls a large language model via the OpenRouter API to parse the input into a structured rules object JSON, passed immediately to the renderer without waiting for the next sensing cycle. The rules object covers ten fields: detail level, layout density, focus mode, font size, colour scheme, summary display, priority display, sidebar visibility, reading assist,

and custom CSS variables, giving the user fine-grained explicit control over interface parameters. Base AI performs implicit inference over the behavioural signals collected by the sensing layer, producing a declarative adaptation decision constrained by the rules object. Its primary inference path is a rule engine that triggers every 60 seconds, and also fires immediately whenever the Settings AI rules object is updated; an optional LLM path is invoked only when the rules have changed significantly, with periodic evaluation handled by the rule engine to conserve API quota. The two engines operate in coordination through a priority arbitration mechanism, ensuring that explicit user preferences always take precedence over behavioural inference. A keyword-matching fallback engine is built into the inference layer, maintaining basic adaptive functionality when the API is unavailable or the network is constrained.

The **renderer** is the only component in DCAF with direct DOM write access, responsible for translating the declarative decision JSON produced by the inference layer into interface changes that users can perceive. Concentrating interface modification rights in a single component eliminates the state management risks that arise when multiple sources manipulate the DOM directly. The renderer operates through four execution channels. The CSS variable channel achieves global hot-updating of visual parameters by dynamically modifying CSS custom properties on the document root, without re-rendering the entire DOM tree. The component library channel maintains a set of pre-registered interface component skeletons; the renderer dynamically inserts corresponding components into the DOM according to the `components` field of the decision JSON, rendering functional elements such as summary panels and priority badges in an AI-safe manner. The visibility control channel dynamically manages the visibility of interface elements according to the `visibility` field, suppressing visual noise when the system infers high cognitive load. The focus mode channel applies `focus-dimmed` and `focus-highlighted` classes to progressively attenuate non-essential areas and foreground the current task focus.

Before each write operation, the renderer computes a difference hash between the incoming decision and the last executed one, discarding the new directive as redundant if the difference falls below a set threshold—preventing interface flickering caused by multiple highly similar directives produced within a short time window. All executed adaptations are recorded in a history stack as complete state snapshots, including the de-

cision content and the corresponding Settings AI rules object at that moment, supporting unlimited undo and redo across the full session with full state restoration on rollback. Each adaptation decision is surfaced to the user through a persistent bottom notification bar displaying the reason field content, with Undo, Redo, and Accept controls and a 20-second auto-accept timeout that does not interrupt the user's current task flow.

4.1.2 Prototype Implementation

The prototype[1] is built using HTML5, CSS3, and JavaScript ES6, running as a single-page application in a modern browser with no additional installation required. It adopts a modular architecture in which three core modules—ResearchLogger, Inference Module, and Renderer—communicate through an event bus rather than direct references, keeping each module independently testable while maintaining a strict one-to-one correspondence with DCAF's three-layer structure. The baseline and adaptive versions of the interface are shown in Figures 4 and 5 respectively; the Settings AI input panel is illustrated in Figure 6; the bottom notification bar in Figure 7; and the activated Focus Mode in Figure 8.

Sensing Module (ResearchLogger) ResearchLogger is mounted on the host application's DOM event system using an event-listener-driven architecture, passively capturing user behaviour without interfering with existing interface logic. Each of the four trackers applies debounce and throttle mechanisms to prevent high-frequency events such as mouse movement from placing undue pressure on the system. Every 10 seconds, ResearchLogger calls each tracker's aggregation method, compressing behavioural events within the sliding window into numerical metrics—mouse entropy, mean switching interval, hover hit count, and reading speed—which are written to the shared context buffer and trigger the inference module's periodic evaluation.

Inference Module The inference module comprises two parallel engines. Settings AI receives natural language input from the user, calls the LLM via the OpenRouter API,

[1]The complete source code for both prototype versions is publicly available at https://github.com/LSP2080/Adaptive-Email-client-Prototype.git.

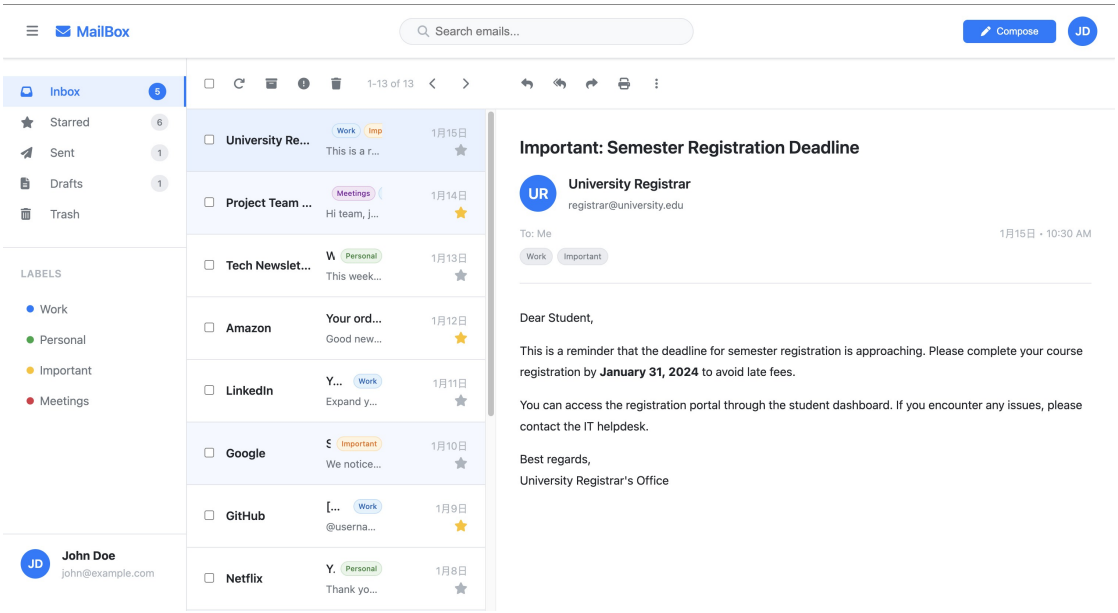


Figure 4 The adaptive email client prototype: baseline version. All AI modules are disabled; the interface remains in its static initial state throughout the session.

and parses the response into a structured rules object JSON covering ten interface parameters. It maintains a conversation history to support multi-turn interaction, and provides four preset configurations (minimal, focused, detailed, relaxed) to lower the barrier to preference expression. When an API call fails, the system automatically falls back to a local keyword-matching engine. Base AI extracts the latest behavioural snapshot from the context buffer at each trigger, constructs a structured prompt combining the rules object and behavioural metrics, and calls the LLM to generate a declarative decision JSON. The decision JSON contains five fields: `cssVars` specifying CSS custom property key-value pairs to modify; `components` declaring the list of functional components to render; `visibility` controlling the display state of interface elements; `focusMode` indicating whether to activate focus mode; and `reason` providing human-readable explanatory text for the notification bar. The two engines are coordinated through a priority arbitration mechanism ensuring that Settings AI rules always function as high-priority constraints within which Base AI's behavioural inference operates.

Renderer The renderer translates incoming decision JSON into interface changes through four execution channels, as described in Section 4.1.1. At the execution level, it

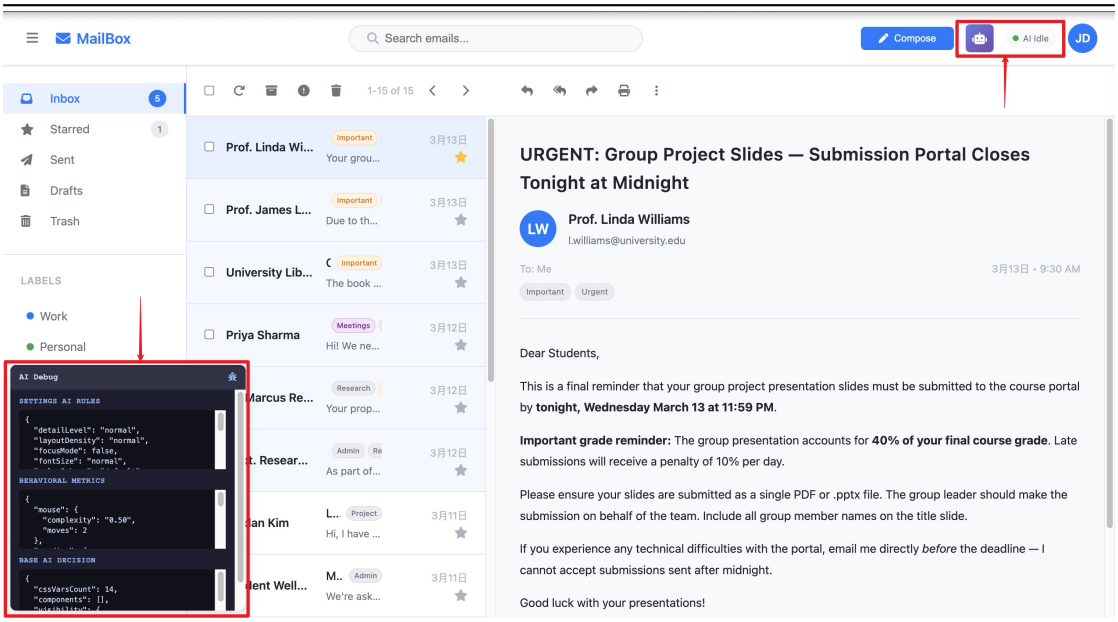


Figure 5 The adaptive email client prototype: adaptive version with the Settings AI input panel open. The user can express interface preferences in natural language at any point during the session.

generates a decision fingerprint before each write operation to filter redundant directives. All executed adaptations are recorded in a *decisionHistory* array with a *decisionIndex* pointer; on undo, the renderer replays the snapshot at *decisionHistory[index-1]* and simultaneously restores the corresponding Settings AI rules, achieving complete state restoration.

Two functionally equivalent versions of the prototype are maintained for the experiment. In baseline mode, ResearchLogger runs continuously to collect data, but the inference module and renderer perform no adaptation—the interface remains in its static initial state, serving as the experimental control condition. In adaptive mode, all three layers operate together. Both versions share identical visual design, email dataset (15 messages across five folders, AI-generated and containing no privacy-sensitive content), and task structure; the only difference is whether the dual-AI pipeline is active, ensuring comparability across conditions.

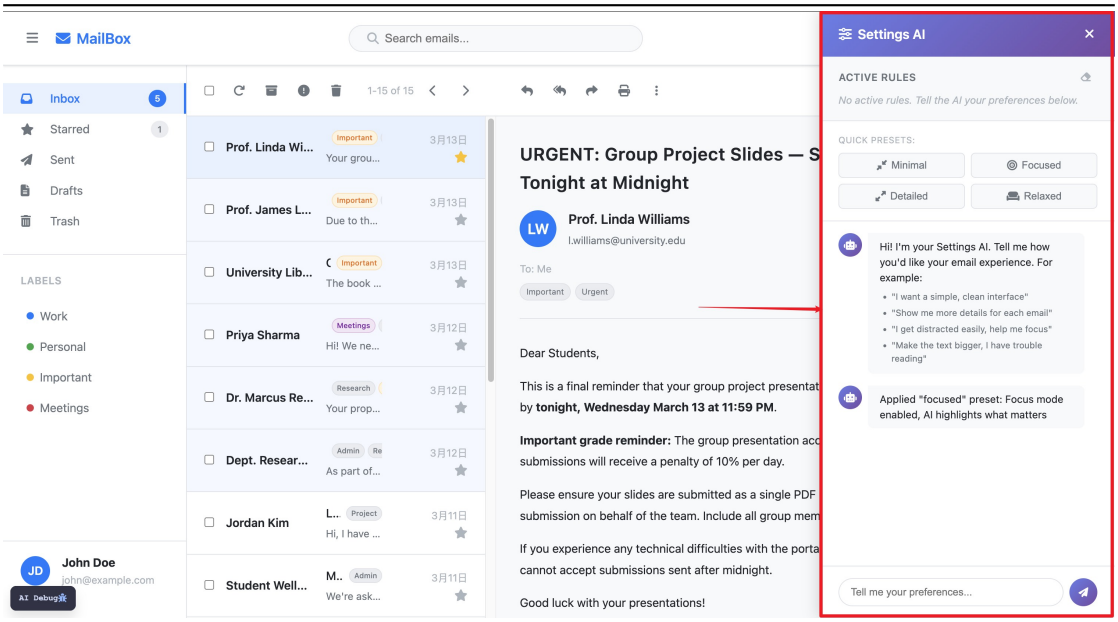


Figure 6 The Settings AI input panel. The user can describe interface preferences in natural language; the panel also provides four preset configurations (minimal, focused, detailed, relaxed) to lower the barrier to preference expression.

4.2 Retrospective Analysis of the First Iteration

4.2.1 First-Generation Framework Architecture

The first-generation framework, as shown before in the Figure 2 in Chapter 3, adopts the same three-layer structure, but the design of each layer differs substantially from the current version of DCAF. The sensing layer continuously and passively collects user interaction behaviour—cursor trajectory, content reading patterns, interface switching frequency, and control hover events—producing a set of numerical behavioural metrics. The inference layer adopts a single-path design, submitting all behavioural metrics to an AI engine that computes two floating-point values—cognitive load and distraction level—through a fixed-weight formula, and maps the result to one of three discrete levels. The renderer transitions the interface to a corresponding preset state based on this classification, with three tiers representing progressively greater degrees of visual simplification and assistive intervention. The inference cycle fires periodically at a fixed interval, recomputing the full state at each trigger and determining whether to switch adaptation levels. The overall process constitutes a unidirectional passive pipeline: the

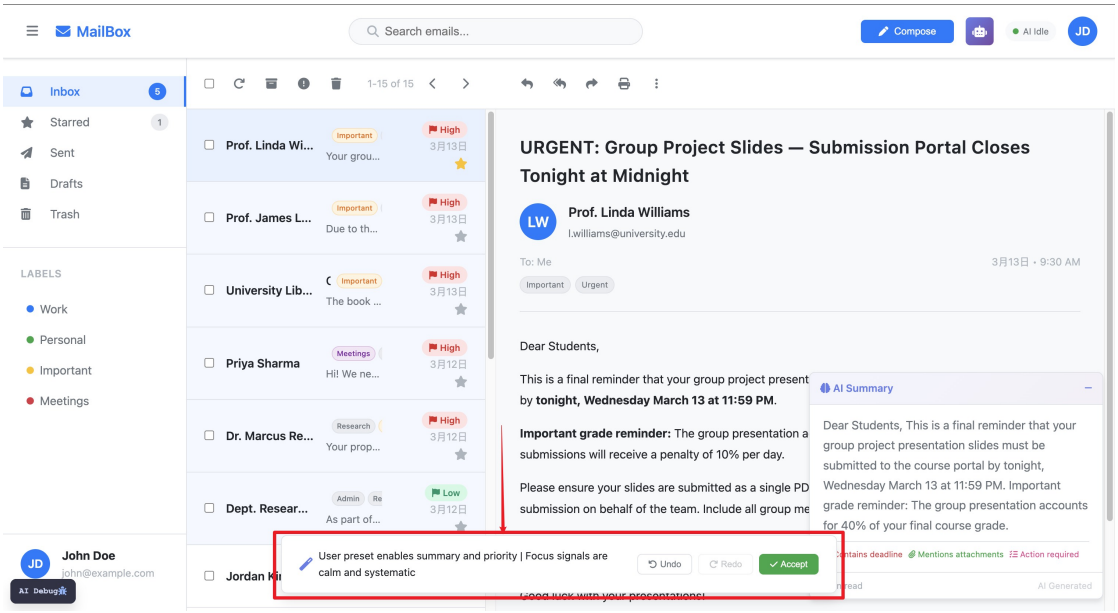


Figure 7 The bottom notification bar surfacing an adaptation decision, displaying the reason for the change alongside Undo, Redo, and Accept controls. A 20-second auto-accept timeout is applied during periods of inactivity.

system makes all adaptation decisions autonomously based on sensed data, with no channel for user intent input.

4.2.2 Structural Limitation Analysis

The first-generation prototype surfaced five categories of structural limitation in practice. These are not isolated implementation issues but point toward systemic constraints at the level of framework design.

Inference fineness. Three discrete levels cannot express the continuous variation of user cognitive state with any nuance. Each evaluation cycle either produces no change or triggers a level jump, resulting in abrupt step-wise interface transitions rather than smooth, incremental responses. The underlying cause is that the inference layer compresses complex cognitive state into three integers, discarding the information carried by intermediate states.

Absence of an explicit user control channel. The first-generation framework contains no mechanism for users to actively express preferences. All adaptation decisions are

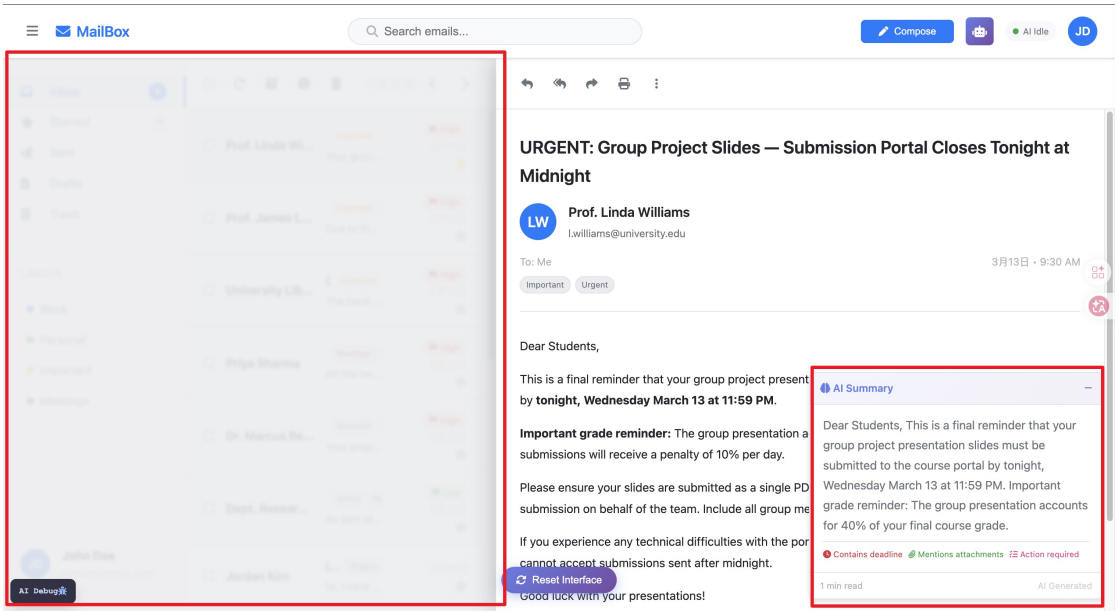


Figure 8 The adaptive email client prototype: Focus Mode activated. Non-essential interface elements are progressively attenuated, with the active reading area foregrounded to reduce visual competition and support sustained attention.

made unilaterally by the system on the basis of sensed data. Users occupy a purely passive position throughout the adaptation process, with no means of communicating intent to the system or intervening in a decision before it is executed.

Interruptive adaptation feedback. Each interface change triggers a modal panel with undo, redo, and accept controls. This notification mechanism forces an interruption to the user's current task at the moment of adaptation, imposing a cognitive cost that partially offsets whatever benefit the adaptation itself may have delivered.

Incomplete undo support. The undo system maintains only a single-step pending state, supporting rollback only of the current unconfirmed change with no history across multiple adaptation cycles. For users who may not notice that an adaptation has occurred until several minutes later, this mechanism offers no practical recourse.

Limited adaptation output vocabulary. The renderer operates exclusively by toggling CSS classes, switching between three coarse-grained preset states. It cannot dynamically introduce functional components or apply fine-grained adjustments to individual visual parameters, confining the expressive range of adaptation to a narrow set of predefined

configurations.

4.2.3 Drive Toward the Second Iteration

These five categories of limitation, together with the user needs evidence drawn from the recruitment questionnaire study, constitute the design inputs for the second iteration. The coarse inference granularity pointed toward replacing discrete level output with a declarative JSON decision structure, allowing fine-grained and composable adaptation directives. The absence of an explicit control channel indicated the need for an independent inference path dedicated to user-expressed preferences. The interruptive feedback mechanism and incomplete undo support revealed systemic insufficiencies in the renderer's user agency design. The limited adaptation output vocabulary identified the need to expand the renderer's execution architecture. Together, these identified problems defined the boundaries within which the second-generation framework was designed.

4.3 Recruitment Survey Results

4.3.1 Participant Background

The survey received 62 complete responses. The sample skewed toward working-age adults: 43 respondents fell between 25 and 44 years of age, 10 were aged 18–24, and the remainder were 45 or older. Gender distribution comprised 35 males, 26 females, and 1 non-binary or preferring not to disclose. Educationally, 33 respondents held a bachelor's or equivalent degree and 29 held a master's degree. Occupationally, the sample was dominated by office workers or professionals (27) and students (14), with the remainder comprising technical workers, researchers, and creative practitioners. Most respondents fell at beginner-to-intermediate levels of technical familiarity (49), with 9 reporting programming or development experience. The majority spent fewer than 30 minutes per day on email (39 respondents), and only 5 reported daily email time exceeding one hour.

Regarding attention-related background, 4 respondents reported a formal diagnosis of an attention-related condition, and 14 reported self-suspected attention difficulties or were in the process of seeking a diagnosis; the remaining 41 reported no such background.

Awareness of attention-related conditions was generally low: 28 respondents had no awareness at all, 12 had heard the term but knew nothing of its symptoms, and only 20 described themselves as having at least a basic understanding.

On AI tool usage, most respondents had some prior experience: 24 were familiar with specific applications such as ChatGPT or Midjourney, 14 had a deeper understanding of AI principles or limitations, and 8 had professional AI experience. In terms of frequency, 16 reported daily or near-daily use and 23 used AI tools a few times per week, with only 5 having never used an AI tool.

4.3.2 User Attitudes and Needs

Data from the 62 survey responses points to a tension that runs through the entire dataset: users hold high expectations for adaptive interfaces, while simultaneously maintaining a clear wariness toward AI systems that make decisions independently. This psychology of wanting adaptation without relinquishing control is perhaps the single most important user need that adaptive system design needs to take seriously.

On attitudes, a clear majority of participants (approximately 85%) responded positively to the idea of interfaces that automatically adjust based on user state—around 40% considered this very useful and wanted to use it immediately, and a further 45% found it somewhat useful and worth trying (Figure 9). Very few worried it might interfere with their usage habits, and no one selected completely unwilling to use it. It should be noted, however, that the sample skewed toward participants with higher technical familiarity and prior AI tool experience; this positive distribution may not generalise to a broader user population.

The picture shifts when the question moves from acceptance to trust. Around half of participants chose rule-based semi-automatic adjustment as their preferred approach, while only around one in ten trusted fully automatic AI-driven adjustment (Figure 10). The gap is not evidence that users reject adaptation—they clearly do not. It suggests instead that the adaptation process itself needs to be legible and predictable; black-box decision-making produces discomfort independent of whether the outcome is correct.

Concerns about loss of control reinforce this reading. When asked about their biggest

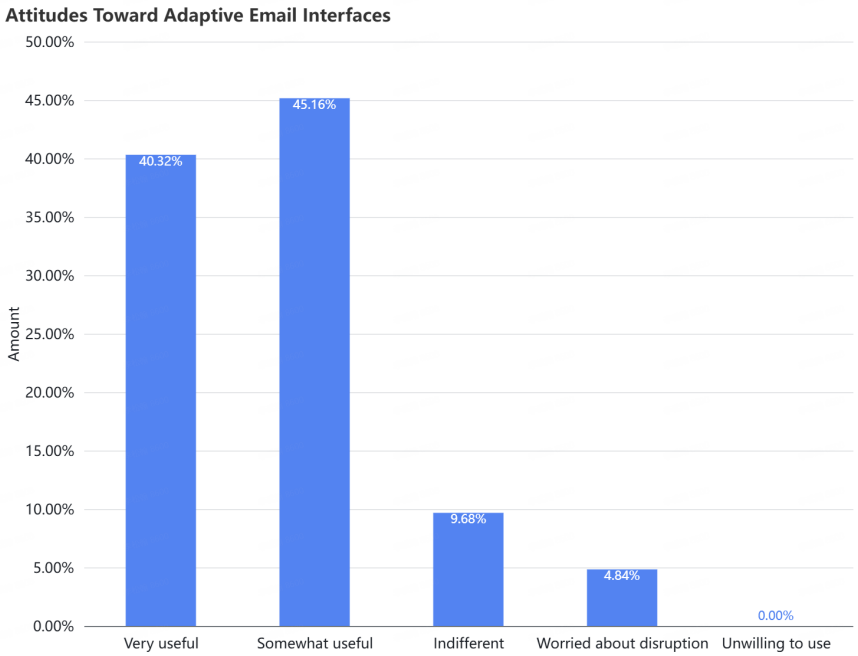


Figure 9 User attitudes toward adaptive email interfaces (N=62).

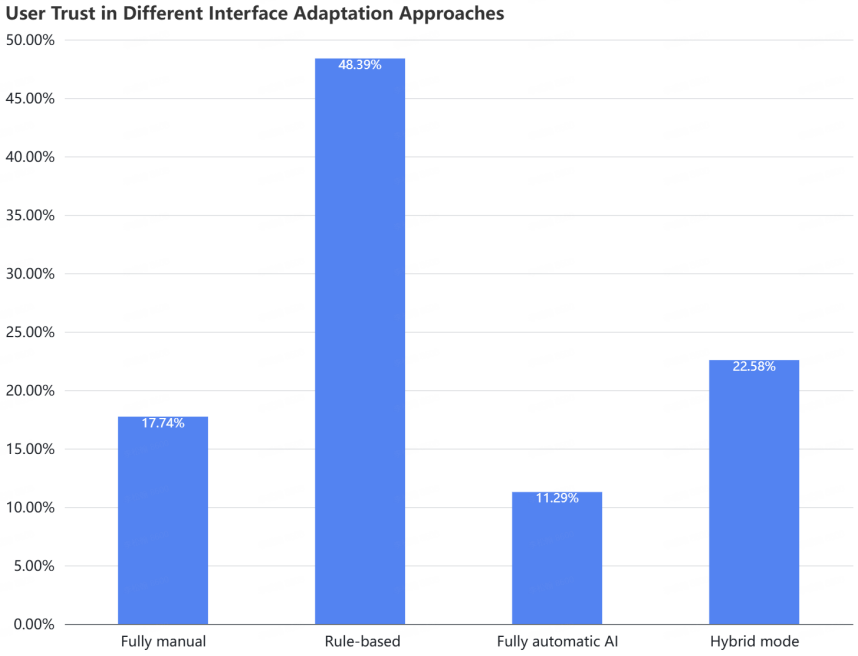


Figure 10 User trust in different interface adaptation approaches (N=62).

worry regarding AI-driven interface adjustment, loss of control ranked third behind privacy concerns and accuracy concerns (Figure 11). One participant noted in the open-response field of the questionnaire: “I hope the system gives me suggestions, but the final decision is mine.”

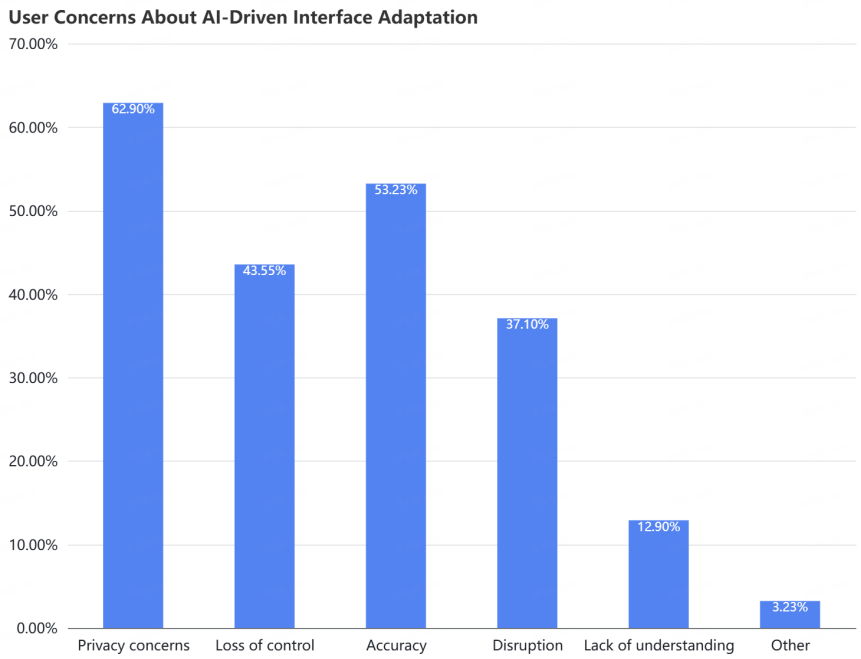


Figure 11 User concerns about AI-driven interface adaptation (N=62, multiple responses allowed).

User acceptance of data collection showed an equally clear hierarchy (Table 2). Behavioural data—reading duration (mean 3.39/5) and mouse movements (3.18/5)—attracted relatively high acceptance, while personally identifiable information scored lowest (2.24/5). Users are not opposed to sharing data with adaptive systems; they are selective about what kind of data, and on the condition that the system remains transparent and reversible.

These findings sit in direct conversation with Roda and Thomas’s [12] core claim about attention-aware systems: that reconciling system autonomy with user agency is the fundamental design challenge. The preference for rule-based over black-box AI adjustment, expressed by close to half of all respondents before any exposure to the system, is substantial enough to treat as an architectural constraint rather than a stylistic preference: when that proportion of users signal rejection of a particular adaptation mechanism at

Acceptance Level	Data Type				
	Mouse movement	Click logs	Reading duration	Email content	Personal information
1 Completely unacceptable	3 (4.8%)	4 (6.5%)	2 (3.2%)	6 (9.7%)	20 (32.3%)
2 Unacceptable	6 (9.7%)	5 (8.1%)	3 (4.8%)	8 (12.9%)	15 (24.2%)
3 Neutral	34 (54.8%)	33 (53.2%)	31 (50.0%)	31 (50.0%)	20 (32.3%)
4 Acceptable	15 (24.2%)	16 (25.8%)	21 (33.9%)	14 (22.6%)	6 (9.7%)
5 Completely acceptable	4 (6.5%)	4 (6.5%)	5 (8.1%)	3 (4.8%)	1 (1.6%)
Mean Score	**3.18**	**3.18**	**3.39**	**3.00**	**2.24**

Table 2 User Acceptance of Different Types of Data Collection (N = 62)

the outset, it marks a boundary condition for system acceptability rather than a matter of individual taste.

In general, these findings served as direct design input for the second iteration. The strong preference for rule-based over fully automatic adjustment informed the decision to ground Base AI's primary inference path in a rule engine rather than relying solely on LLM calls. The consistent demand for transparency and reversibility shaped the renderer's notification and history stack design. The selective but conditional acceptance of behavioural data collection confirmed that passive sensing was viable, provided the system's use of that data remained visible and undoable. In this sense, the recruitment survey functioned not only as a participant screening instrument but as a lightweight needs elicitation study whose outputs were directly traceable to specific architectural decisions in DCAF's second-generation design.

4.4 User Study Results

4.4.1 User Study Participants

Ten participants were selected from the 62 survey respondents on the basis of three criteria: diversity of attention profiles (ensuring a reasonable mix of participants with and without attention-related experiences), range of technical familiarity (to avoid over-representing technically expert users), and acceptance of behavioural data collection (participants who indicated high sensitivity in the survey were excluded).

The ten participants (P1–P10) comprised six males and four females, ranging in age from

20 to 41 years ($M = 26.4$). All held at least an undergraduate degree; three were doctoral or postgraduate students and the remainder held bachelor's or master's qualifications. Seven were studying or working in technology-adjacent fields (computer science, HCI, software engineering, or UX design), while three came from non-technical backgrounds (education, business, and creative industries).

Participants were categorised into two groups on the basis of their attention profile. The attention-related group ($n = 6$; P1, P3, P5, P6, P7, P8) included five participants with a formal clinical diagnosis and one with a self-reported mild or suspected attention difficulty. Diagnosed participants had received their diagnoses between one and eight years prior to the study; four managed their condition without medication at the time of the study, and one used medication intermittently. The general group ($n = 4$; P2, P4, P9, P10) comprised participants who reported no current attention-related diagnosis or suspicion, though P9 noted occasional difficulties with sustained attention in demanding contexts without meeting any clinical threshold.

All participants reported using email regularly for academic or professional purposes, though daily volume varied considerably: estimates ranged from fewer than 10 emails per day (P4, P10) to 50 or more (P2, P6). Prior experience with AI writing or productivity tools was common across the group; eight of the ten participants reported using at least one AI tool at least weekly, consistent with the broader survey sample. Table 3 summarises participant characteristics.

Nine of the ten participants (all except P10) took part in semi-structured follow-up interviews lasting 30 to 50 minutes. P10 was unable to participate in the interview phase due to time constraints. While session logs and questionnaire data capture behavioural patterns and ratings, they cannot reveal how participants understood the adaptive features, how they reasoned about specific design decisions, or how they responded to particular moments during the session; the interviews were included to address precisely this layer of the data [66].

ID	Age	Gender	Background	Field	Attention	Order
P1	24	M	Computer Science	PGT	C	BF
P2	28	M	Business	Industry	G	DF
P3	22	F	HCI	PGT	C	BF
P4	31	M	Software Eng.	Industry	G	DF
P5	23	F	Psychology	UG	C	BF
P6	25	M	Computer Science	PGT	C	BF
P7	27	M	UX Design	Industry	S	DF
P8	20	F	Education	UG	C	BF
P9	41	M	Creative Ind.	Industry	S	DF
P10	33	F	Business	Industry	G	DF

Table 3 Participant characteristics (N = 10). Attention status: C = confirmed diagnosis; S = self-reported or suspected; G = general (no diagnosis). Condition order: BF = baseline first; DF = dual-AI first.

4.4.2 Quantitative Analysis

Faster and Less Lost: Task Performance Under Dual-AI Assistance

Across the ten participants, total task completion time was consistently lower in the dual-AI condition than in the baseline. The baseline condition comprised five structured email tasks covering information retrieval, priority ranking, email composition, deadline management, and decision making. The mean reduction was 27.4% ($SD = 9.8\%$), from a baseline mean of 1,144.8 seconds to a dual-AI mean of 831.4 seconds. The pattern held for every participant without exception, though the magnitude varied considerably: P3 achieved the largest gain (40.9% reduction, from 1,320 s to 780 s), followed closely by P6 (38.9%, from 1,750 s to 1,070 s), while P2 recorded the smallest improvement (10.7%, from 1,597 s to 1,426 s). Figure 12 presents per-participant task times across both conditions.

Participants who started from higher baseline times—those who tended to navigate between emails repeatedly, re-read the same messages, or spend prolonged time on the priority-ranking and deadline-management tasks—showed the steepest gains under dual-AI assistance. P3 and P6, both participants with confirmed attention-related diagnoses, exhibited exactly this pattern: their baseline sessions involved a high number of re-orientations, and the adaptive focus mode and AI summaries appear to have most

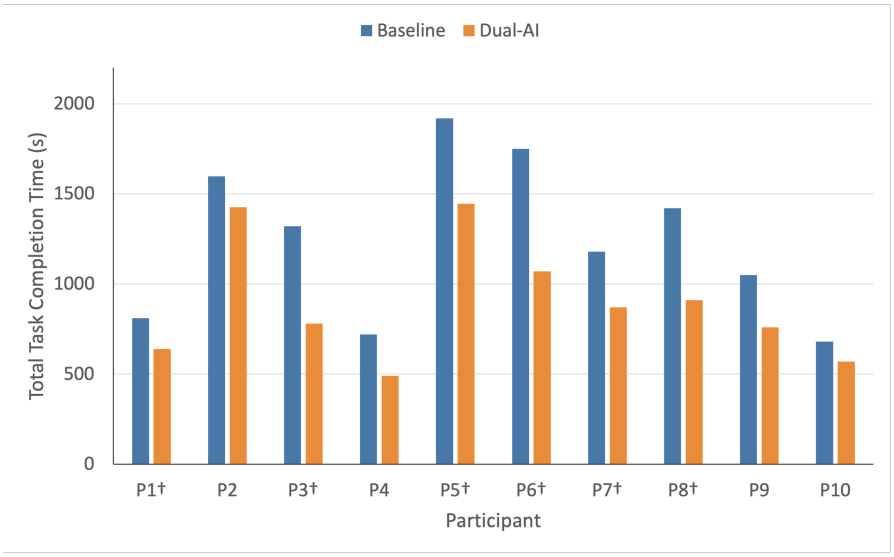


Figure 12 Per-participant total task completion time (seconds) under baseline and dual-AI conditions (N = 10).

directly addressed the disorganised browsing behaviour that was driving their longer times. P2, the outlier with the smallest time reduction, maintained consistently long task times in both conditions and will be discussed further in Section 4.4.3.

The session logs offer three additional indicators that corroborate the time reduction data and help explain its source.

Email opening events. For clarity, this metric counts the total number of times any email was opened during a session; reopening the same message counts as a separate event. The number of email opening events per session fell by a mean of 35.8% in the dual-AI condition (baseline $M = 39.1$, dual-AI $M = 25.1$). This reduction does not simply reflect fewer tasks being completed—all participants completed all five tasks in both conditions. The more plausible explanation is that the AI summary cards reduced the need to reopen the same email multiple times to retrieve information. P6 recorded 52 email opening events in the baseline session compared to 28 in the dual-AI condition (a 46.2% reduction), while P5 went from 48 to 38. In contrast, P10 and P4, who began from lower baseline event counts (24 and 26 respectively), showed proportionally smaller reductions (33.3% and 26.9%). Interview data corroborates this interpretation: participants with attention-related difficulties reported reopening emails because they could not recall what a message contained—a behaviour that the summary feature appeared to displace

(see Section 4.4.3). Figure 13 presents per-participant email opening event counts across both conditions.

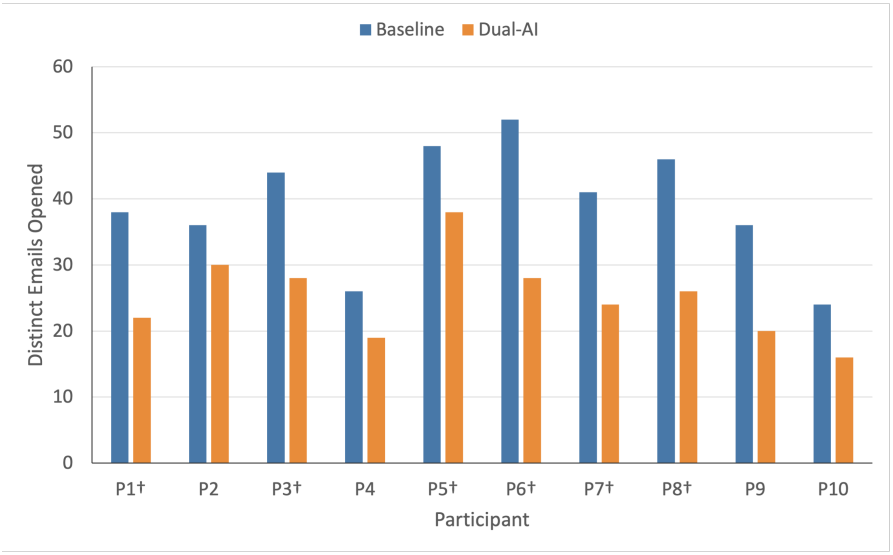


Figure 13 Email opening events per session under baseline and dual-AI conditions (N = 10). Each reopening of the same message is counted as a separate event.

Mouse movement complexity. Mouse movement complexity—measured as the ratio of directional changes to total cursor distance, a proxy for unfocused or searching navigation—fell by a mean of 19.6% in the dual-AI condition (baseline $M = 0.118$, dual-AI $M = 0.095$). The participants who showed the steepest complexity reductions were also those with the largest time gains: P3 dropped from a complexity score of 0.142 to 0.108, and P6 from 0.138 to 0.098, suggesting that the interface adaptations reduced the amount of purposeless cursor movement associated with trying to locate relevant content. P10 and P2, both general-group participants who started from lower baseline complexity scores, showed the most modest reductions (0.095 to 0.079 and 0.085 to 0.079, respectively), consistent with the interpretation that the adaptive features primarily benefit users whose baseline navigation is already more erratic. Figure 14 presents per-participant complexity scores across both conditions.

Average reading speed. Average reading speed (words per minute, WPM) showed a mean decrease from 293.2 WPM in the baseline to 205.1 WPM in the dual-AI condition (Figure 15). P8 illustrates what this shift looks like in practice: moving from 348 WPM in the baseline—a rate consistent with skim-reading—to 182 WPM in the dual-AI con-

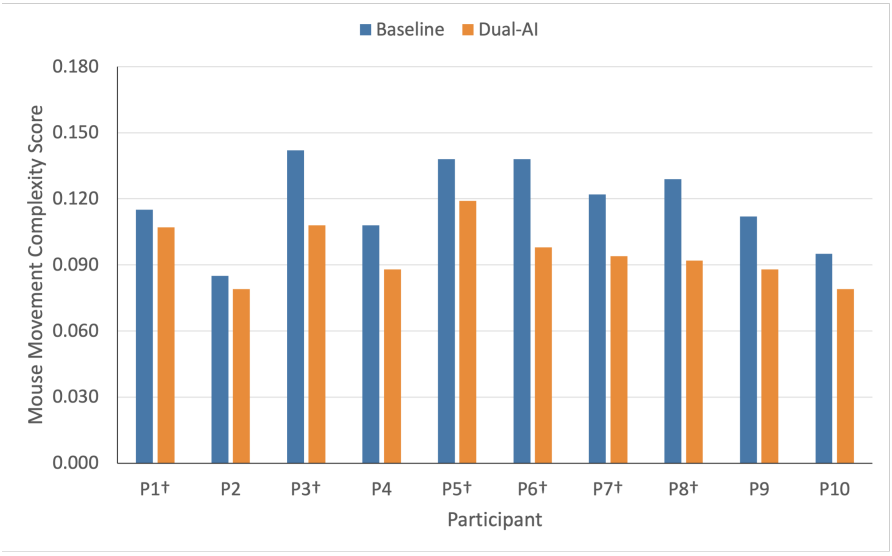


Figure 14 Per-participant mouse movement complexity scores (directional changes / total cursor distance) under baseline and dual-AI conditions (N = 10). Lower scores indicate more directed, purposeful navigation.

dition, while also completing tasks more quickly overall. This pattern suggests that AI summaries shifted the reading mode from indiscriminate scanning toward targeted verification of specific details, with the slower WPM reflecting deeper, more deliberate engagement rather than reduced efficiency. P4 (248 to 315 WPM) and P5 (168 to 195 WPM) present a different pattern: both showed increased WPM alongside reduced task times, suggesting these participants used the summaries to identify which emails to skip entirely, reading fewer messages but more quickly. Whether reduced WPM reflects deeper attention or a different reading strategy is a question the current data cannot resolve with certainty; what is clear is that the dual-AI condition consistently changed how participants engaged with the inbox, and in every case this change was accompanied by faster overall task completion.

Taken together, the three behavioural indicators point to a coherent picture: the dual-AI client shortened task times primarily by reducing the re-navigational overhead that accompanies information retrieval under working memory pressure. Participants recorded fewer email opening events, moved their cursors less erratically, and completed the task set in less time. The effect was largest for those who started from a position of greatest difficulty.

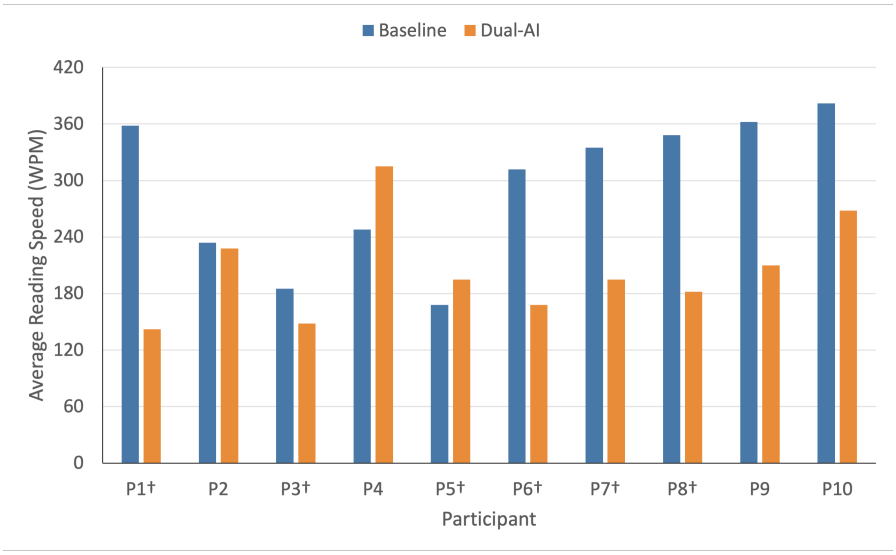


Figure 15 Per-participant average reading speed (words per minute) under baseline and dual-AI conditions (N = 10).

The System is Usable, Even for Those It's Designed to Help Most

The System Usability Scale yielded a mean score of 74.8 ($SD = 8.0$, $n = 10$) for the dual-AI condition, placing the system in the "good" range according to Bangor et al.'s adjective rating scale [67], and above the commonly cited acceptability threshold of 68 [1]. The system crossed this threshold not only at the group level but for the majority of individual participants: six of the ten scores exceeded 70, and eight of ten exceeded the 68-point acceptable boundary. Figure 16 presents individual SUS scores for both groups.

The two participants whose SUS scores fell below the threshold were both from the attention-related group. P5 scored 67.5 and P6 scored 65.0—the lowest individual score in the dataset. P6's result is worth examining in context: this participant showed the most erratic baseline navigation (mouse complexity score of 0.138) and the highest number of email opening events in the baseline condition (52). Their dual-AI session showed the second-largest time reduction in the dataset (38.9%), confirming that the system altered their interaction behaviour substantially. The below-threshold SUS score therefore does not indicate that the system was unhelpful to P6; it suggests that the adaptation cycle was still too disruptive to generate a strong sense of usability fluency, even as it was measurably changing their behaviour in a positive direction. P6 noted in the

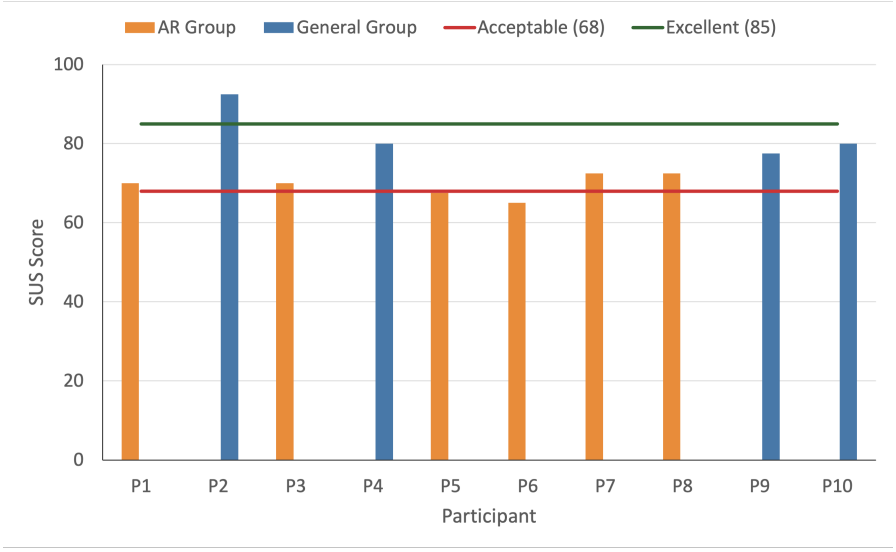


Figure 16 Individual SUS scores for all ten participants under the dual-AI condition. Participants are grouped by attention profile: AR = attention-related group ($n = 6$); G = general group ($n = 4$). The horizontal line indicates the acceptability threshold of 68 [1].

open-response field accompanying the SUS questionnaire that the interface adjustments sometimes felt "a bit abrupt" and that they were "not always sure whether something had changed or whether it was me." The perceptual discontinuity that comes with poorly-signalled automatic adaptation is a known driver of reduced SUS scores in adaptive systems [13], and these results suggest DCAF has not yet fully resolved it.

The attention-related group ($n = 6$) returned a mean SUS of 69.6 ($SD = 2.9$), compared to 82.5 ($SD = 6.8$) for the general group ($n = 4$), as summarised in Table 4. These figures should be interpreted with caution: with a sample of this size, a single participant's score can shift a group mean considerably, and the two groups were not counterbalanced. Consistent with the exploratory positioning of this study, no statistical inferences are drawn. The observed difference nonetheless points to a pattern worth investigating in future work: participants with cognitive accessibility needs appear to have found the system somewhat less fluent to use, even when it produced the largest objective performance gains for them.

The NASA-TLX data provide a complementary picture of subjective cognitive load, with per-participant scores shown in Figure 17. The overall mean was 57.4 ($SD =$

	SUS		NASA-TLX	
Group	**Mean**	**SD**	**Mean**	**SD**
AR ($n = 6$)	69.6	2.9	61.1	3.8
G ($n = 4$)	82.5	6.8	51.8	10.4
Overall	74.8	8.0	57.4	8.2

Table 4 SUS and NASA-TLX scores by participant group. AR = attention-related group; G = general group.

8.2), which falls below the range typically associated with cognitive overload in task-focused HCI studies [2]. Given the sample size, this observation should be treated as a preliminary trend rather than a generalisable finding. The attention-related group's mean was somewhat higher (61.1, $SD = 3.8$) than the general group's (51.8, $SD = 10.4$). The wider spread in the general group is visible at the individual level: P2 found the session demanding (NASA-TLX = 67.0), while P10 and P4 found it quite light (44.2 and 46.0 respectively).

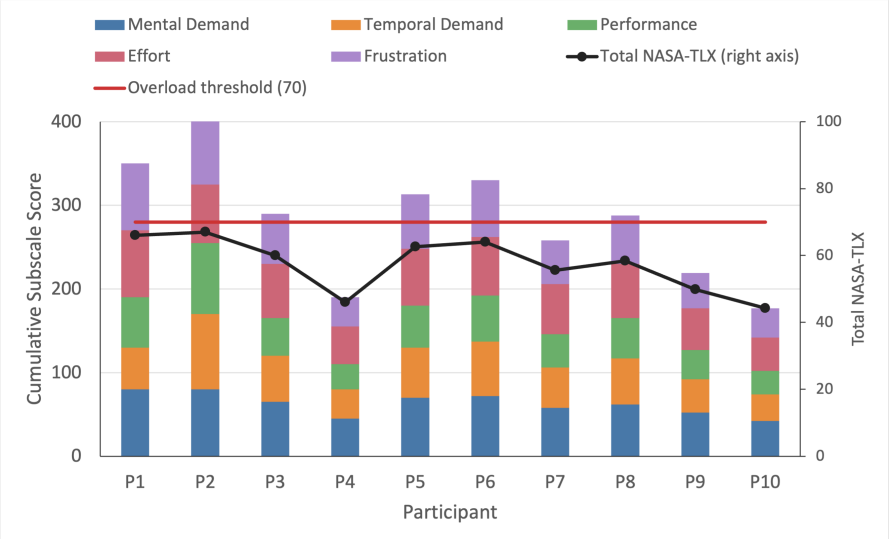


Figure 17 Individual NASA-TLX scores for all ten participants under the dual-AI condition. The horizontal line indicates the approximate overload threshold of 70 [2].

The most analytically interesting case in this dataset is P2, who recorded both the highest NASA-TLX score (67.0) and the highest SUS score (92.5). P2's baseline task time was by far the longest in the dataset (1,597 s), suggesting a methodical working style that

was not fundamentally altered by the dual-AI system—hence the smallest time reduction (10.7%) and persistently high NASA scores across both conditions. P2 also showed the lowest mouse complexity and a relatively modest email opening event count in the baseline, indicating that their extended times came from careful deliberation rather than disorganised navigation. For this participant, the adaptive features did not address a navigation problem they did not have; instead, they appear to have been appreciated at a qualitative level—for the summaries, the importance labels, and the structural clarity—while engaging with a system that behaved differently from a standard email client added to rather than reduced the cognitive load. This profile—high appreciation, unchanged work pace, elevated load—may represent a class of users for whom adaptive interfaces deliver perceived value without behavioural efficiency gains, a distinction the current design does not explicitly account for. Table 5 presents individual scores for reference.

ID	Attention	Group	SUS	NASA-TLX
P1	C	AR	70.0	66.0
P2	G	G	92.5	67.0
P3	C	AR	70.0	60.0
P4	G	G	80.0	46.0
P5	C	AR	67.5†	62.6
P6	C	AR	65.0†	64.0
P7	S	AR	72.5	55.6
P8	C	AR	72.5	58.4
P9	S	G	77.0	49.8
P10	G	G	80.0	44.2
AR Mean			69.6	61.1
G Mean			82.5	51.8
Overall			74.8	57.4

Table 5 Individual SUS and NASA-TLX scores under the dual-AI condition. Attention status: C = confirmed; S = suspected; G = general. SUS scores below 68 are marked with †.

Overall, the SUS and NASA-TLX results suggest that the dual-AI adaptive client is a usable system for the population it targets, at least within this small exploratory sample. The residual usability deficit observed in P5 and P6 points more specifically to the disruptive quality of silent, abrupt adaptation—a design problem that the qualitative analysis in Section 4.4.3 addresses in detail.

4.4.3 Qualitative Results

Tell Me What Changed, and Let Me Undo It

In all nine interviews, transparency was central to how participants evaluated the adaptive system. A consistent position emerged independently across sessions: participants were not opposed to the system changing the interface, but the majority were quite firmly opposed to it doing so without informing them. This concern appeared in the first interview and recurred in every subsequent one. All interview participants are identified by their experimental identifiers (P1–P9); P10 did not take part in the interview phase.

The most fundamental demand was the right to post-hoc awareness: knowing what had changed after an adaptation occurred. P1 stated that "any adjustment needs to be reported afterwards, even small changes are not acceptable if executed silently." P4 expressed the same requirement in more practical terms: he would read a change description before deciding whether to accept it, and the 20-second auto-dismiss window felt "maybe a bit too fast." Both positions rest on the same logic: if a user has not noticed a change, they cannot meaningfully control it—regardless of whether an accept/reject dialogue technically exists.

It is worth addressing directly why DCAF was designed to adapt automatically rather than asking the user first. Evidence from the first iteration cycle suggested that frequent confirmation requests introduce their own cognitive burden, particularly for users with attention-related difficulties; and the recruitment survey data indicated high acceptance of the adaptive concept in principle. DCAF therefore adopted an "adapt first, notify after, always reversible" strategy rather than an "ask first, then adapt" approach. The interview data both validate this direction—participants accepted automatic adaptation—and identify where its current implementation falls short: the notification mechanism is not yet adequate.

The clearest account of what this means for the target population came from P5, a participant with a confirmed attention-related diagnosis. After a prototype demonstration, she noticed that the system had already made several adjustments she had not perceived. Her reaction was immediate: "What did it just do? What specifically changed?" When the researcher pointed to the status indicator in the top-right corner, she replied that she

had not noticed it. “For someone like me, I need a more direct way of telling me what changed—not asking me find it, but making it come to me.” She later described the problem as follows: a user with attention difficulties “might not notice that the AI did three things in the last minute; by the time I noticed the interface has changed, several minutes may have passed.” Standard single-step undo is therefore not enough; the system needs to support multi-step reversal covering the full session history, because the user’s awareness of any single change cannot be assumed. P5 proposed a lightweight change-log panel—modelled on VS Code’s Git changes sidebar—that would keep recent adaptations persistently visible without interrupting the current task.

The reversibility requirement was equally consistent, though it took different forms depending on each participant’s context. P2 described undo as a “last-resort control mechanism,” something whose mere presence was reassuring even if rarely used. P3 proposed a layered consent model: the first instance of any given adaptation type would require explicit acknowledgement, but subsequent instances could proceed by default. P6, coming from a professional context where email management carries accountability, stated three explicit conditions: adaptations must be “reversible, logged, and dismissible at any time.” His objection to interrupt-style pop-ups was equally direct: “Don’t pop something up at me while I’m in the middle of an email.” He advocated instead for a passive status-bar indicator, with detail available on hover.

The contrast between P5’s position—who needed salient, proactive notification—and P6’s—who rejected any form of interruption—is not a contradiction to be resolved so much as a meaningful signal in itself. It suggests that no single notification mechanism will suit all users, and that the system needs to allow users to configure the degree of notification intrusiveness rather than imposing a fixed mode. This has a direct design implication: a tiered notification system, in which users can set their own threshold between proactive alerts and passive indicators, may be necessary to accommodate the range of preferences represented in this data.

P7, drawing on her UX design background, identified the underlying principle behind these preferences as progressive disclosure: information about what the system has done should exist at the periphery of the user’s attention rather than demanding it. Progressive disclosure was not an explicit design principle in DCAF’s initial conception, but P7’s framing reveals a gap between user expectations and the current implementation, and

points toward a direction for subsequent design iterations. She proposed a contrast view—analogous to Figma's version-history comparison feature—letting users see the before-and-after state of any adaptation rather than just a textual description. Her design-level conclusion was that the change-log should not be a buried settings page but a first-class UI element, persistent and low-friction to reach.

These accounts converge on a single underlying requirement: user agency over an adaptive system. Transparency and reversibility are two aspects of this same requirement—users need to be able to know what the system has done and to undo it, and the adequacy of each depends on the other. A visible change log with no undo mechanism is information without recourse; a one-step undo with no change log cannot be used by someone who did not notice a change had occurred. The current prototype provides a single notification prompt and a one-step accept/reject mechanism, as shown before in the Figure 7,—a starting point, but one that falls short of the multi-step, always-accessible, low-interruption transparency architecture that the interview data consistently calls for.

Not All Adaptations Are Equal: A Priority Layer of Feature Value

If transparency refers to how the system should be adjusted—keeping users informed and in control—then another equally important question is which aspects of the system should be adjusted first. Interview data suggests that not all DCAF features are considered equally valuable, and the credibility of a system largely depends on whether its interventions are targeting the right issues.

The feature value hierarchy reported here was not elicited through a formal ranking task. It emerged naturally during the semi-structured interviews: when participants were asked which features they found most useful and which they could do without, eight of the nine interviewees spontaneously organised their responses into a consistent priority order without being prompted to rank. The resulting hierarchy was strikingly uniform despite significant differences in participants' backgrounds and attention profiles.

The highest-value feature, identified unprompted by the majority of participants, was the AI-generated email summary. P1 named content summary and priority ordering as the two functions he would retain if forced to choose, explicitly describing them as the core of what makes an adaptive client useful. P3, the developer participant, stated that the

summary function had the highest priority among all features, and that it was the one adaptation he would actively seek out even if the rest of the system were unavailable. P5's evaluation was the most emphatic: this one feature alone, she said, was enough to justify using the system. Her spontaneous ranking placed email summaries first, and when she elaborated her reasoning, she grounded it directly in her own experience: the summary card displaces the need to open the same email multiple times to recover forgotten information, a behaviour pattern she identified as one of her primary sources of email-related cognitive load. P6, approaching from a professional rather than a clinical angle, converged on the same conclusion: his first two desired features were intelligent priority marking and email thread summarisation, both of which he categorised as tools for processing information, as distinct from tools for changing the interface. The distinction implicit in this framing—between content-level intelligence and display-level adjustment—proved to be the organising principle of the entire hierarchy.

Focus Mode, the feature that hides the sidebar and simplifies the email list to reduce visual competition, occupied a consistent second tier, as shown before in the Figure 8. Its reception was more differentiated than the summary feature, with participants who had attention-related difficulties placing it substantially higher than general users. P5 ranked it second in her spontaneous list, noting that it directly addresses the multi-panel disorientation that characterises her experience with standard email clients. P4, who identified mild attention tendencies, described the feature as interesting but noted that the 60-second trigger interval was too fast, producing transitions that felt intrusive rather than supportive. His proposed alternative was a longer detection window, closer to five minutes, combined with a smoother visual transition, so that the mode change would feel like a gradual settling into focus rather than an abrupt restructuring. P1, who does not have an attention-related diagnosis, was less enthusiastic: he found the sidebar-blurring effect not particularly useful, and appreciated the mode's existence primarily as a manual override option rather than as an AI-triggered adaptation. This pattern—attention-related participants placing Focus Mode near the top of the value ranking while general users treated it as secondary—was consistent across the interview set and aligns with the design intent of the feature.

The lowest tier comprised visual micro-adjustments: automatic changes to background colour, font size, and layout proportions. P3 accepted visual layout changes as manage-

able but was emphatic that email ordering and content modification were categorically different and unacceptable. P1 noted that he did not want font size to increase automatically, though he accepted the existence of a reading assistance mode as an option. The clearest structural articulation of this tier came from P7, who proposed an explicit three-level taxonomy: content intelligence, covering email summaries and priority recommendations, at the top; focus assistance, covering Focus Mode and sidebar control, in the middle; and visual fine-tuning, covering colours, typography, and layout proportions, at the bottom. Her prescription followed directly: the system's resources should be concentrated on the first tier, and low-value functions like colour adjustment should be optional choices the user makes, not default automatic behaviours. The design implication she drew was that the trigger threshold for each tier should differ: high-value features can justify more active AI intervention, while low-value features should require explicit user opt-in.

This hierarchy carries a practical consequence for DCAF's design. The current system treats all four adaptation types—layout change, priority reordering, focus mode activation, and visual adjustment—as outputs of the same inference cycle, triggered at the same frequency and governed by the same confirmation mechanism. The interview data suggest this conflates features with different user value profiles, different trust requirements, and different appropriate trigger frequencies. A system that reserved its automatic adaptation behaviour for high-value features while relegating visual micro-adjustments to a user-configurable preset layer would be likely to generate less friction and more trust than one that treats all adaptations as equally deserving of AI initiative.

4.4.4 Summary of Analysis

On objective performance and subjective experience, the dual-AI condition was associated with meaningful reductions in task completion time (-27.4%), mouse navigation complexity (-19.6%), and redundant email interactions (-35.8%), alongside lower self-reported cognitive load (NASA-TLX mean: 57.4) and acceptable-to-good usability ratings (SUS mean: 74.8). These gains were observed across both general users (SUS: 82.5) and those with attention-related difficulties (SUS: 69.6), though the gap between subgroups suggests room for further optimisation for users with cognitive accessibility

needs.

Transparency and reversibility emerged as universal prerequisites for user acceptance. Before participants were willing to delegate adaptation decisions to the system, they required a persistent, non-intrusive change log and multi-step undo capability. This was not an idiosyncratic preference: it was a consistent demand across all nine interviews.

Feature value and privacy boundaries were drawn consistently along a content-versus-behaviour axis. Email summarisation and focus assistance were ranked as the highest-value features, while content-layer AI processing was accepted only when locally executed and explicitly user-initiated. In other words, participants were willing to let AI help them understand information, but maintained a clear boundary around AI touching content itself without their knowledge.

These findings are taken up in the following chapter, where this thesis discuss design implications, the limitations of the current prototype, and directions for future work.

5 Discussion

5.1 Research Questions Revisited

RQ1: Feasibility of DCAF Instantiation and Framework Applicability

From a technical standpoint, the three-layer architecture of DCAF was fully instantiated in this research. The sensing layer's four trackers continuously and passively collected behavioural signals; the dual-AI inference layer successfully handled two qualitatively different types of input—explicit user preferences and implicit behavioural inference; and the renderer translated declarative decisions into perceptible interface changes through a four-channel execution architecture. The core sense–infer–adapt pipeline operated reliably under experimental conditions, and the keyword-matching fallback engine maintained basic adaptive functionality when the API was unavailable, demonstrating a degree of robustness under network-constrained conditions. In this respect, the answer to RQ1 is affirmative: all core components of DCAF can be operationalised into a functional prototype.

Technical instantiability, however, is only one dimension of feasibility. The more consequential question is for whom the framework is suited, and under what conditions. The results suggest that DCAF produces its most pronounced objective effects for users operating under elevated cognitive load and exhibiting more disorganised baseline navigation—a profile particularly concentrated among participants with attention-related difficulties. This does not mean the framework is exclusively useful for users with clinical diagnoses; the general group also benefited, albeit with smaller effect magnitudes. A more precise characterisation is that DCAF's value correlates with the degree of cognitive pressure a user faces under baseline conditions: the higher the baseline difficulty, the greater the marginal benefit of adaptation.

On applicable domains, this study instantiated DCAF in an email client, but the three-layer structure is inherently scenario-agnostic. The sensing layer's trackers can be adapted to any web-based application; the dual-AI inference architecture does not depend on a specific task type; and the renderer's four-channel execution mechanism is transferable to other information-intensive interfaces. In principle, any context involving sustained

information processing and decision-making under cognitive load—document editing, code review, online learning platforms—satisfies the basic conditions for DCAF deployment. The effectiveness of sensing signals and the adequacy of inference logic in different domains would nonetheless require independent validation, a point taken up further in Section 5.3.

RQ2: Objective Performance Outcomes

The objective performance data present a consistent positive pattern. Task completion time decreased by a mean of 27.4%, email opening events fell by 35.8%, mouse movement complexity dropped by 19.6%, and average reading speed decreased from 293.2 WPM to 205.1 WPM. These four indicators converge on a single explanatory mechanism: the dual-AI system reduced the overall cost of information retrieval by alleviating the repetitive navigation behaviour that accompanies working memory pressure. The AI summary feature reduced the need to reopen the same email repeatedly; focus mode reduced purposeless cursor movement; together, they acted on task completion time through complementary pathways.

It should be noted that these results come from a small-sample exploratory study and should not be interpreted as statistically generalisable conclusions. The directionality of effect sizes is consistent, but individual variation is considerable—P2's time reduction was 10.7%, while P3's reached 40.9%. This variation is itself informative: it suggests that DCAF's effects are strongly conditioned by users' baseline behavioural profiles, rather than producing uniform benefits across the population.

RQ3: Subjective Experience and User Perception

The subjective data present a picture that is internally consistent but carries a notable tension. A mean SUS score of 74.8 places the system in the "good" usability range, and a mean NASA-TLX score of 57.4, below the cognitive overload threshold, suggests that the dual-AI system did not introduce significant additional cognitive burden. The gap between the attention-related group's mean SUS (69.6) and the general group's (82.5), however, creates an interesting counterpoint to the objective data: the group that

showed the largest objective performance gains reported comparatively lower subjective usability.

The qualitative analysis points to the source of this tension: the perceived fluency of automatic adaptation. Users accepted adaptation in principle, but expressed discomfort at the absence of sufficient perceptual control over when and how it occurred. Transparency and reversibility appeared in all nine interviews not as desirable enhancements but as prerequisite conditions for trust. The implication extends beyond DCAF's current implementation: it points toward a broader design principle that the tension between users' need for a sense of control and a system's capacity for autonomous decision-making cannot be resolved simply by improving adaptation accuracy. How the system communicates what it has done, and how readily users can undo it, may matter as much as whether the adaptation decision was correct.

5.2 Design Implications

Transparency and Reversibility as Design Prerequisites

One of the most consistent findings of this study is that user acceptance of adaptive systems has its limits. Across all nine interviews, transparency and reversibility recurred as prerequisite conditions rather than optional enhancements. The implication for adaptive system design is fairly direct: before users are willing to delegate interface decisions to a system, two baseline conditions must first be satisfied—the user must be able to know what has changed, and the user must be able to undo any change they did not want.

These two conditions are mutually dependent in ways that matter for implementation. A system that maintains a complete undo history but remains silent about what changed offers no practical recourse to a user who did not notice a change had occurred; a system that surfaces every change but cannot reverse it turns transparency into informed helplessness. P5's case is particularly illustrative: she only became aware, after a demonstration, that the system had already made several adjustments she had not perceived—pointing to the notification mechanism as a structural factor in the trust-building process rather than a peripheral design detail. Transparency and reversibility need to be conceived and implemented as an integrated whole; treating them as separate feature modules that can

be developed independently tends to produce gaps at the level of user experience.

The Necessity of Human Initiative in AI-Driven Systems

Recruitment survey data showed that close to half of respondents expressed reservations about fully automatic AI-driven adaptation before any exposure to the system, preferring rule-based semi-automatic adjustment instead. The interview data elaborated this preference more precisely: users did not reject AI involvement, but held a clear expectation about who should hold final decision authority. AI provides suggestions and support; the human retains the power to decide. This was the framing that recurred across interviews, regardless of participants' technical backgrounds.

As AI systems continue to grow in capability, this question becomes more pressing. Higher adaptation accuracy also means users find it increasingly difficult to anticipate and understand the system's decision logic—a somewhat counterintuitive tension in which greater AI capability may actually heighten rather than reduce users' need for control. The design space for human initiative should therefore not narrow as AI capability increases; it needs to be more carefully designed. The presence of Settings AI in this framework is itself a response to this principle: even when Base AI can produce reasonably accurate behavioural inferences, users still need a channel through which they can actively express preferences and override system decisions at any point. This is an institutional guarantee of human agency in human–AI collaboration, independent of how capable the AI component happens to be.

A Tiered Triggering Mechanism for Adaptation Features

The interview data revealed a dimension the current DCAF design does not sufficiently address: different categories of adaptive feature carry substantially different value weights and trust thresholds in users' evaluations. AI-generated email summaries and priority labelling were consistently placed at the highest value tier, with users showing the greatest acceptance of AI initiative for this category of content intelligence. Focus mode occupied a middle tier, with acceptance varying considerably by attention profile. Visual micro-adjustments—font size, colour scheme, layout proportions—fell at the lowest tier,

with most users explicitly stating they did not want this category triggered automatically.

The current system treats all four adaptation output types as products of the same inference cycle, triggered at the same frequency and governed by the same confirmation mechanism—implicitly assuming that all adaptation types carry equivalent user value and equivalent triggering legitimacy. The interview data refutes this assumption. A more principled design would establish a tiered triggering mechanism: high-value features in the content intelligence category can be AI-initiated; mid-tier features such as focus assistance can be AI-suggested but require user confirmation before execution; low-value features such as visual fine-tuning should default to opt-out from automatic triggering, existing only as user-configurable presets. This tiering would concentrate the system's AI initiative where it is most likely to generate value, with a corresponding reduction in the adaptation friction users experience.

5.3 Limitations

Several limitations in the design and execution of this study warrant consideration when interpreting the results.

Sample size is the most immediate constraint. Ten experimental participants is consistent with the conventions of early-stage validation in design science research, but it means that no quantitative result carries inferential statistical weight, and group-level comparisons require particular caution—the mean scores of both the attention-related and general groups are each susceptible to considerable shift from a single participant's data. The quantitative findings should be understood as directional trends rather than generalisable conclusions; their value lies in providing effect size estimates and hypothesis generation for subsequent larger-scale research.

Single-scenario validation is equally worth acknowledging. DCAF was validated exclusively within an email client context. Although the three-layer structure is designed to be scenario-agnostic, the validity of sensing signals is highly task-dependent—mouse movement complexity and switching frequency are meaningful cognitive load proxies in an email processing context, but their interpretive power may not transfer equally to other domains. Before deploying DCAF in different application areas, both the signal

selection of the sensing layer and the inference logic of the inference layer would require independent validation.

Behavioural indicators as proxies for cognitive state represent an inherent methodological constraint. Mouse movement complexity, reading speed, and switching frequency are all indirect measures, and the correspondence between each and actual cognitive load is modulated by task type, individual habits, and user differences. This study employed no direct measurement instruments such as eye-tracking or physiological sensors, meaning that the signals collected by the sensing layer can only serve as approximate estimates of cognitive state. P2's case illustrates the practical impact of this limitation clearly: their extended task times arose from careful deliberation rather than disorganised navigation, and the current sensing mechanism cannot reliably distinguish between the two.

Dependence on an external LLM API constitutes a structural limitation on framework reliability. Although a keyword-matching fallback engine was implemented, the inference quality available in degraded mode falls considerably short of normal API operation. Under conditions of network constraint, API interruption, or high response latency, the system's adaptive capacity degrades substantially. Beyond availability, LLM inference is inherently non-deterministic—identical inputs can produce different outputs across calls—introducing challenges for the predictability of system behaviour that the current architecture does not fully resolve.

5.4 Ethical Considerations

This research touches on several ethical dimensions in its design and execution, some of which extend beyond the scope of a single study and carry relevance for broader deployments of the DCAF framework.

Data collection and user privacy represent the most immediate ethical concern. DCAF's sensing layer continuously and passively collects mouse trajectory, reading behaviour, and interface interaction patterns. In this study, all sensing data were processed locally and not stored persistently beyond the session. In wider deployment contexts, however, the storage, transmission, and use of sensing data would directly bear on users' privacy

interests. Recruitment survey data indicated that users show higher acceptance of behavioural data than of personally identifiable information, but this acceptance rests on the condition that the system remains transparent and its use of data remains bounded. If sensing data were used for purposes beyond immediate interface adaptation—such as cross-session behavioural profiling—this condition would no longer hold, and the resulting privacy risks would need to be explicitly constrained in any deployment specification for the framework.

The computational cost and environmental footprint of LLM API calls is a less commonly discussed but non-trivial concern. DCAF's inference layer relies on an external LLM API for semantic reasoning; in a continuously running production environment, high-frequency API calls would generate considerable computational cost and energy consumption. As adaptive systems proliferate across more application contexts, their collective environmental footprint becomes a design responsibility that warrants serious attention. The current implementation does not resolve this tension—both Settings AI and Base AI invoke external API calls under their respective trigger conditions, and the cumulative cost of sustained deployment remains an open question. More systematic approaches to reducing inference overhead are explored as part of the future work outlined in the following section.

The transparency ethics of AI-driven interface decisions raises a deeper issue: when an interface is adjusted autonomously by AI, can users genuinely understand what the system is doing and why. P6's description of being uncertain whether the interface had changed or whether their own perception had shifted reveals that automatic adaptation can introduce subtle disruptions to users' perceptual grounding. For users with cognitive accessibility needs, the impact of such disruption may be more pronounced. The interpretability of AI decisions is therefore not only an engineering design goal but an ethical requirement for protecting users' cognitive autonomy.

The responsibility of inclusive design requires that designers continuously examine whether a framework's service boundaries are inadvertently creating new forms of exclusion, even as adaptive capability is being pursued. The results of this study show that DCAF produced its largest objective gains for the attention-related group, yet this same group returned comparatively lower subjective usability scores. A system that is statistically effective for its target population does not necessarily mean that population

feels respected and supported in using it. The goal of inclusive design is not only functional accessibility but also the preservation of dignity and autonomy at the level of lived experience.

Several of the ethical issues identified here—in particular those concerning privacy protection and environmental cost—have candidate solution directions currently being explored, and are taken up in the following section on future work.

5.5 Future Work

This research represents an initial validation of the DCAF framework, and leaves several directions open for subsequent work.

Revisiting the timing of adaptive triggers is the most immediate engineering-level improvement. The current 60-second periodic cycle for Base AI was set for experimental convenience rather than on the basis of any systematic study of real-world usage rhythms. In practical deployment, continuous periodic adaptation may introduce unnecessary disruption, and users' cognitive state changes are not uniformly distributed across time. Future work should explore more principled triggering strategies—such as event-driven triggers tied to task phase transitions, dynamic cycle adjustment informed by individual usage history, or the introduction of explicit do-not-disturb time windows—to strike a better balance between adaptation responsiveness and continuity of user experience.

Larger-sample validation studies are a necessary condition for translating the exploratory findings of this research into generalisable conclusions. The current ten-participant sample cannot support statistical inference, and group-level comparisons lack adequate statistical power. Subsequent research should expand the sample size, introduce more rigorous experimental controls, and replicate the evaluation of DCAF's effects across user populations with diverse demographic backgrounds and levels of technical familiarity, in order to assess the external validity of the current findings.

Personalisation of the renderer's notification mechanism is the design direction most directly indicated by the qualitative results. Interview data revealed substantial individual variation in notification preferences—P5 required proactive, salient change indicators, while P6 explicitly rejected any form of interruption. The current single bottom notifica-

tion bar cannot simultaneously accommodate both needs. Future work should explore a configurable tiered notification system that allows users to set their own threshold between proactive alerts and passive indicators according to their individual habits, extending the principle of user agency to the design of the notification mechanism itself.

Model localisation represents a fundamental direction for simultaneously addressing privacy protection, offline availability, and environmental cost. The current framework depends on an external LLM API, meaning that users' behavioural data must leave the local device to complete inference, and the system degrades to rule engine mode when the network is unavailable. As lightweight local language models continue to mature— including small-parameter models capable of running on-device—the technical conditions for running the inference layer entirely on the local device are steadily improving. Local inference would eliminate the privacy risk of behavioural data transmission at source, reduce dependence on network connectivity, and substantially decrease the operational cost and environmental footprint associated with API calls.

Cross-scenario and cross-platform migration validation is the critical step toward DCAF realising its declared value as a general-purpose framework. This research confined the framework to a web-based desktop email client, but DCAF's design intent is scenario-agnostic and platform-portable. Future work should extend the framework to mobile applications and toward system-level cross-application adaptive contexts—analogous to the mode in which system-level intelligent assistants such as Siri sense user state at the operating system layer and adjust interaction experiences across applications. This extension involves diversifying sensing signal sources (touch patterns, voice input, application-switching behaviour), developing the inference layer's capacity to integrate cross-application context, and adapting the renderer's execution mechanisms to different UI frameworks. Validating DCAF's portability across a broader platform ecosystem would represent the transition from framework feasibility to framework generalisability.

6 Conclusion

Digital interface design has long faced a problem that remains incompletely resolved: how to sense a user's cognitive state in real time without specialised hardware, and respond in a manner that is transparent and reversible—while placing users with cognitive accessibility needs at the centre of the design rather than treating them as an afterthought. Existing adaptive interface frameworks exhibit limitations across all three of these dimensions: they rely on static user profiles, lack adequate user control mechanisms, or treat the needs of general users and accessibility users as separate problems to be addressed independently.

This thesis proposed and preliminarily validated the Dynamic Context-Aware Adaptive Framework (DCAF), arriving at its current form through two iterative design cycles. Beginning from a literature-driven first-generation framework, and driven by the dual inputs of a parallel user recruitment study and a systematic analysis of first-generation prototype limitations, the research produced a framework incorporating a dual-AI inference architecture and a four-channel renderer. The framework was instantiated as an adaptive email client and evaluated through a within-subject controlled experiment with ten participants and nine semi-structured interviews.

The results indicate that the DCAF-powered dual-AI adaptive client produced positive outcomes across both objective performance and subjective experience dimensions. Task completion time decreased by a mean of 27.4%, email opening events fell by 35.8%, mouse movement complexity dropped by 19.6%, and the System Usability Scale mean reached 74.8. Effects were most pronounced among participants with attention-related difficulties, though general users also benefited. Qualitative analysis revealed that transparency and reversibility function as prerequisite conditions for user acceptance of adaptive systems rather than optional enhancements, and that users' evaluations of different adaptation features follow a consistent value hierarchy in which content intelligence functions receive the highest endorsement.

The contributions of this research operate at two levels. At the framework level, DCAF provides a transferable architecture that cleanly separates sensing, inference, and rendering responsibilities and communicates through standardised interfaces, offering a reusable design specification for building cognitively-aware adaptive systems across dif-

ferent application contexts. At the knowledge level, this research empirically confirms the necessity of a mixed-initiative architecture in adaptive systems, and elevates transparency and reversibility from design recommendations to architecture-level constraints supported by user data.

The exploratory nature of this research defines the boundaries of its conclusions. Small sample size, a single application scenario, and dependence on an external API all constitute limitations that subsequent work will need to address. Nevertheless, the framework, prototype, and evaluation methods established in this paper provide ideas and directions for larger-scale, multi-scenario verification. Adaptive interfaces as a research field stand at a critical juncture between technical feasibility and general usability, and DCAF is one step moving in that direction.

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A Questionnaire

Adaptive Email Client User Research Recruitment Questionnaire

Thank you for your interest in this research. This study aims to evaluate an adaptive email client designed for users with different attention patterns and explore the feasibility of AI-assisted interface adaptation. This questionnaire will take approximately 8–10 minutes to complete. All information will be kept strictly confidential and used only for research purposes.

1. What is your age range? [Single Choice] *

- Under 18
- 18–24
- 25–34
- 35–44
- 45–54
- 55–64
- 65 or above

2. What is your gender? [Single Choice] *

- Male
- Female
- Non-binary / Prefer not to say

3. What is your occupation? [Single Choice] *
If you select 'Other', please specify.

- Student
- Office worker / Professional
- Researcher / Academic
- Technical staff / Programmer
- Creative / Design industry
- Other _____

4. What is your educational background? [Single Choice] *

- High school or below
- College / Bachelor's degree
- Master's degree
- PhD or above

5. How familiar are you with technology? [Single Choice] *

- Beginner (basic usage)
- Intermediate (daily use, can solve common issues)
- Advanced (understand technical principles)
- Expert (programming / development experience)

6. Approximately how much time do you spend on email (work + personal combined) per day? [Single Choice] *

- Less than 30 minutes
- 30 minutes – 1 hour
- 1–2 hours
- 2–4 hours
- More than 4 hours

7. Which email client do you mainly use? [Single Choice] *
If you select 'Other', please specify.

- Gmail
- Outlook
- Apple Mail
- NetEase Mail / QQ Mail
- Enterprise email system (customized)
- Other _____

8. On average, how many emails do you receive per day? [Single Choice] *

- Fewer than 10
- 10–30
- 30–50
- 50–100
- More than 100

9. How do you handle unread emails? [Single Choice] *

- I must read all of them to feel at ease
- I only read important ones and ignore the rest
- I mark them for later but often forget
- I let them pile up and clean them in batches occasionally

10. What are your main email usage scenarios? [Multiple Choice] *

- Work communication
- Academic communication
- Personal matters
- Subscriptions / newsletters
- Shopping / bills
- Social notifications
- Other

11. Do you use email categorization features (folders / labels / filters)? [Single Choice] *

- Never
- Occasionally
- Frequently
- Heavily rely on them with a well-established system

12. How do you deal with long emails? [Single Choice] *

- Read them all at once
- Read them in several sessions and often don't finish
- Only read the beginning and the end
- Save them for later but rarely revisit them

13. Have you ever been diagnosed with or suspected that you may have attention-related issues (e.g., ADHD)? [Single Choice] *

- Yes, formally diagnosed
- Yes, self-suspected / seeking diagnosis
- No
- Prefer not to say

14. Please specify the diagnosis date: [Open-ended] *
(Only shown if Q13 = formally diagnosed)

- _____

15. If diagnosed, what is your current situation? [Single Choice] *
(Only shown if Q13 = formally diagnosed)

- Untreated / not taking medication
- Receiving treatment (therapy / counseling)
- Currently taking medication
- Stopped medication / recovered

16. How familiar are you with ADHD? [Single Choice] *

- Not familiar at all
- Heard of it but do not know the symptoms
- Basic understanding (core symptoms: inattention, hyperactivity, impulsivity)
- Fairly familiar (different types, comorbidities, etc.)
- Very familiar (self/family member has ADHD or studied it professionally)

17. In your opinion, how does ADHD mainly affect daily work and life? [Multiple Choice] *
If you select 'Other', please specify.

- Difficulty sustaining attention
- Easily distracted / zoning out
- Severe procrastination
- Difficulty managing time
- Emotional regulation problems
- Social difficulties
- Low work/study efficiency
- Other _____

18. Do you think technology products should be specially designed for people with ADHD/attention disorders? [Single Choice] *

- Strongly agree
- Agree
- Neutral
- Disagree
- Strongly disagree
- Not sure

19. Have you used any apps or tools designed to help with focus? (e.g., Forest, Focusmate, Pomodoro timers) [Single Choice] *

- Never used
- Tried but did not continue
- Occasionally use
- Frequently use
- Rely on them regularly

20. How familiar are you with artificial intelligence (AI)? [Single Choice] *

- Not familiar at all
- Heard of the basic concept
- Know some applications (e.g., ChatGPT, Midjourney)
- Fairly familiar (understand principles and limitations)
- Very familiar (professional background / development experience)

21. How frequently do you use AI tools in daily life? [Single Choice] *

- Never
- Occasionally (1–2 times per week)
- Frequently (3–5 times per week)

- Very frequently (multiple times per day)

22. If you use AI, what are the main purposes? [Multiple Choice] *

- Writing assistance (emails, articles)
- Information search / summarization
- Programming assistance
- Translation
- Creative generation
- Learning support
- Entertainment / casual chat

23. How much do you trust AI in the following areas? [Matrix Scale Question]

- Medical diagnosis
- Autonomous driving
- Content recommendations (shopping/videos)
- Automatic interface adaptation
- Personal assistant / scheduling
- Automatic email replies

24. If an email client could automatically adapt its interface based on your state (e.g., simplified layout, larger fonts, highlighted key points), you would find it: [Single Choice] *

- Very useful, would like to use it immediately
- Somewhat useful, willing to try
- Indifferent
- Concerned it may interfere with my habits
- Would not want to use it at all

25. What is your biggest concern about 'AI automatically adapting interfaces'? [Multiple Choice] *
If you select 'Other', please specify.

- Privacy concerns (AI collecting behavioral data)
- Loss of control (system makes decisions for me)
- Accuracy (AI makes incorrect judgments)
- Disruption / annoyance (changes too frequently)
- Lack of transparency (not understanding why changes happen)
- Other _____

26. Which type of interface adaptation do you trust more? [Single Choice] *

- Fully manual settings (I adjust everything myself)
- Rule-based semi-automatic (If A then B, predictable)
- Fully AI-automatic (AI learns my habits and adjusts automatically)
- Hybrid mode (AI suggests, I confirm)

27. How acceptable are the following types of data collection to you? [Matrix Scale Question]

- Mouse movement tracking
- Click/interaction logs
- Reading duration
- Email content (for summarization)
- Personal identity information

28. How would you like the system to handle your data? [Single Choice] *

- Fully processed locally, no cloud upload
- Upload anonymized data for improvement
- Upload allowed but usage must be clearly disclosed
- Doesn't matter, I trust technology companies

29. Have you ever stopped using an app or service due to privacy concerns? [Single Choice] *

- Never
- Occasionally
- Frequently
- I always consider privacy issues

30. Would you be willing to participate in a 60–90 minute online/offline research session? [Single Choice] *

- Yes
- No

31. If yes, which time slots are most suitable for you? [Multiple Choice]

- Weekday mornings (9:00–12:00)
- Weekday afternoons (13:00–17:00)
- Weekday evenings (17:00–20:00)
- Weekends

32. Which participation method do you prefer? [Single Choice]

- In-person (suitable for the Uppsala, Sweden area)
- Online remote (Zoom/Teams)
- Either is fine

33. If you are interested in participating, please leave your contact information (email/phone): [Open-ended]

- _____

B Interview Guide

DCAF USER STUDY

Interview Script

Semi-structured interview protocol for the adaptive email client prototype

PROJECT	DCAF — Adaptive Email Client Prototype
FORMAT	Semi-structured interview
DURATION	~45–55 min total
PARTICIPANTS	P1 = ADHD user | P2 = Neurotypical user
VERSION	v1.0 — Internal Research Use Only

LEGEND

`[R]`	Researcher script — say these exact words
`[Stage direction]`	Action or setup instruction (not spoken)
`→ Probes`	Follow-up if response is shallow
`— TRANSITION —`	Bridge phrase between sections
`[If silent]`	Alternative if participant doesn't respond

PRE-INTERVIEW CHECKLIST

- ☐ **Consent** Obtain written or verbal consent from participant
- ☐ **Recording** Ask permission to audio-record the session
- ☐ **Prototype** Have both BaseCase_test and Dual-AI Version_test loaded in browser
- ☐ **Set email** Know which email set (A or B) participant is assigned
- ☐ **Notes** Prepare paper notes sheet (Researcher Notes Template)
- ☐ **Phone** Phones on silent for both researcher and participant
- ☐ **Time** Remind participant of ~45 min duration

0 Introduction & Warm-up

BOTH PARTICIPANTS · ~5 MIN

◆ *Settle the participant. Offer water. Start audio recording after consent.*

Opening statement

RESEARCHER

"Hi, thank you so much for coming today. My name is [name], and I'm a researcher working on adaptive interfaces. Today's session is about how people interact with email — and whether an AI system that automatically adapts the interface could be helpful. There are absolutely no right or wrong answers. I'm genuinely interested in your honest thoughts and experiences, even if they're critical. Before we begin — do you have any questions? And are you okay with me recording the audio?"

◆ *Wait for consent. Start recorder. Note start time.*

Warm-up questions

These questions are low-stakes — just to get the participant talking comfortably. Don't press too hard; move on after ~60 seconds per question.

QW 1

RESEARCHER

"Can you tell me a little bit about your daily routine — what devices and apps you typically use?"

→ PROBES

- *Do you spend more time on a laptop or phone?*
- *What's the first app you open in the morning?*

IF SILENT: *'For example, are you mostly on a laptop, or do you do most things on your phone?'*

QW 2

RESEARCHER

"And how much time do you spend on email each day, roughly?"

→ PROBES

- *Do you check it at specific times or whenever a notification comes in?*
- *Do you ever feel like email takes up too much of your day?*

→ TRANSITION

Great, that's really helpful context. Now I'd like to learn a bit more about your personal experience — particularly around focus and digital tools.

1A ADHD Experience

P1 (ADHD PARTICIPANT) ONLY · ~10 MIN

⚠ **SKIP THIS SECTION FOR P2. Go directly to Section 1B.**

◆ *Adopt a relaxed, non-clinical tone. Avoid 'symptoms'. Use 'experience', 'challenges', 'differences'.*

Q 1

RESEARCHER

"I'd love to start with your experience living with ADHD day-to-day. How would you describe it in your own words?"

→ PROBES

- *When did you first become aware that you had ADHD?*
- *Has it changed how you approach studying or work?*
- *Are there days when it feels more manageable than others?*

IF SILENT: *'It doesn't have to be formal — even just one or two words that come to mind.'*

Q 2

RESEARCHER

"Are there specific digital activities — like reading long emails, browsing, or multitasking across apps — where ADHD makes things noticeably harder?"

→ PROBES

- *What tends to happen — do you get distracted, or does it take much longer?*
- *Is reading a long email thread especially challenging?*
- *What about notifications — helpful or overwhelming?*

Q 3

RESEARCHER

"Do you notice differences in your ability to focus depending on the time of day, or the environment you're in?"

→ PROBES

- *Is there a time of day when you feel most productive?*
- *Does background noise help or hurt?*
- *What's your ideal setup when you need to really focus?*

Q 4

RESEARCHER

"Have you ever changed any interface settings — font size, colour, notifications, layout — specifically to help you focus?"

→ PROBES

- *What did you change? Did it actually help?*
- *Do you use dark mode, or any browser extensions?*
- *Have you ever used accessibility features like high contrast or text-to-speech?*

Q 5

RESEARCHER

"Thinking specifically about email — is there anything about standard email interfaces that you find particularly frustrating or distracting?"

→ PROBES

- *Too many visual elements on screen at once?*
- *Is the inbox list stressful to look at?*
- *Do long email threads feel hard to follow?*

→ TRANSITION

Thank you for sharing all of that — it really helps me understand your experience. I'd now like to zoom out a bit and talk about digital interfaces more generally.

1B ADHD Awareness

P2 (NEUROTYPICAL PARTICIPANT) ONLY · ~5 MIN

⚠ SKIP THIS SECTION FOR P1. Go directly to Section 1A, then proceed to Section 2.

◆ *Keep this brief and non-leading. Don't explain ADHD in detail — you're gauging their existing understanding.*

Q 1

RESEARCHER

"Before we look at the prototype, I'd like to understand your general awareness of ADHD. How would you describe what you know about it?"

→ PROBES

- *Is your knowledge mostly from media, or from personal interaction?*
- *Have you ever worked closely with someone who has ADHD?*

Q 2

RESEARCHER

"If you have worked with or known someone with ADHD, did you notice any differences in how they used digital tools or managed information?"

→ PROBES

- *Did they seem overwhelmed by interfaces that felt normal to you?*
- *Did they develop any workarounds or personal systems?*

IF SILENT: *'Even if you haven't — what would you imagine might be different for them?'*

Q 3

RESEARCHER

"Do you think standard software and interface design considers neurodiverse users like people with ADHD?"

→ PROBES

- *What do you think is missing?*
- *Have you ever personally benefited from accessibility features?*

→ TRANSITION

Great. Now let's talk a bit more broadly about digital interfaces and any frustrations you personally have with them.

2 Daily Interface Pain Points

BOTH PARTICIPANTS · ~10 MIN

◆ *Do NOT mention the prototype yet. Let them surface their own pain points naturally.*

Q 1

RESEARCHER

"Thinking about apps or websites you use every day — is there anything that consistently annoys or slows you down?"

→ PROBES

- *Is there a specific UI pattern — like a popup, a cluttered menu, or a confusing layout — that comes to mind?*
- *Does it happen more with work tools, or personal apps?*

IF SILENT: *'Even something small — like an animation that's too slow, or a button in the wrong place.'*

Q 2

RESEARCHER

"Have you ever felt that an interface was just too 'busy' — too much information at once, to the point where you felt overwhelmed?"

→ PROBES

- *What did you do — close it, push through, or give up entirely?*
- *Did you find a way to simplify it, or did you just live with it?*
- *Does email inbox feel this way sometimes?*

Q 3

RESEARCHER

"On the flip side — has an interface ever felt too sparse or unhelpful when you actually needed guidance?"

→ PROBES

- *For example, when you were trying to learn a new app or complete an unfamiliar task?*
- *What would have helped?*

Q 4

RESEARCHER

"When you're tired, stressed, or under time pressure — does the way you interact with technology change?"

→ PROBES

- *Do you avoid certain apps or tasks?*
- *Does it take you longer to find things or make decisions?*
- *Do you make more mistakes or misclick more?*

Q 5

RESEARCHER

"Have you ever personalised or customised an interface — changing settings, using browser extensions, or enabling accessibility features?"

→ PROBES

- *Walk me through what you changed and why.*
- *Did you discover these features yourself, or did someone tell you about them?*
- *Did the changes actually help?*

→ TRANSITION

That gives me a really clear picture. Let's shift gears slightly — I'd like to talk about AI, and how you feel about it being involved in your work.

3 Views on AI Collaboration

BOTH PARTICIPANTS · ~10 MIN

Q 1

RESEARCHER

"Do you currently use any AI-powered tools in your daily life or work? Things like ChatGPT, Copilot, Grammarly, or even smart autocomplete?"

→ PROBES

- *How often, and for what kinds of tasks?*
- *Has any of it actually changed how you work?*
- *Are there any AI tools you've tried and given up on?*

Q 2

RESEARCHER

"How do you generally feel about AI making decisions or suggestions on your behalf — even small ones, like reordering your inbox?"

→ PROBES

- *Does it depend on the stakes — is it fine for small things but not important decisions?*
- *Do you worry about losing control, or does it feel like a relief?*
- *Has your feeling about this changed over time?*

Q 3

RESEARCHER

"Can you think of a specific situation where AI collaboration genuinely helped — saved you time, reduced stress, or made something easier?"

→ PROBES

- *What was it about that interaction that made it feel good?*
- *Did you trust the AI's output immediately, or did you check it?*

IF SILENT: *'Even a small thing counts — like autocomplete finishing your sentence perfectly.'*

Q 4

RESEARCHER

"Have you ever felt that an AI crossed a line — did something without your knowledge or permission that felt intrusive or wrong?"

→ PROBES

- *How did it affect your trust in that tool?*
- *Did you stop using it, or adjust your settings?*

Q 5

RESEARCHER

"If an AI could learn your working style over time — noticing when you're focused, when you're distracted, how fast you typically read — what would you most want it to do with that

information?"

→ PROBES
- *Organise things for you? Filter noise? Change the visual layout?*
- *Would you want to know what it's learned about you?*
- *Are there things you'd want it to stay out of?*

→ TRANSITION
This is a perfect lead-in to what I'd like to show you next. Let me pull up the prototype.

4 Prototype Interaction & Reaction

BOTH PARTICIPANTS · ~10 MIN

◆ *Open the assigned prototype (BaseCase_test OR Dual-AI Version_test) in the browser. Sit slightly to the side — participant should feel like they're driving, not being watched.*

4.1 Free exploration

RESEARCHER
"I'm going to show you an email client prototype we've been building. I'd like you to just explore it for a few minutes — there's no task yet, just click around and see what you find. Think aloud if you can — just narrate whatever you're noticing or thinking."

◆ *Allow 2–3 minutes of free exploration. Take notes on where they click, what they linger on, any verbal reactions.*

Q 1

RESEARCHER
"What's your first impression? What stood out to you?"

→ PROBES
- *Does the layout feel familiar or different from email clients you use?*
- *Is there anything that felt confusing or unclear?*

4.2 Triggering an adaptation

◆ *FOR DUAL-AI VERSION: Trigger an adaptation via the Settings AI panel (e.g., type 'I'm finding it hard to focus — make the interface simpler'). Watch participant's face as the UI changes. FOR BASELINE VERSION: Skip this sub-section.*

RESEARCHER
"I'm going to type something into this chat panel here. Watch what happens to the interface."

Q 2

RESEARCHER
"Did you notice anything change? What did you think when you saw that?"

→ PROBES
- *Was it surprising, or did you expect something like that?*
- *Did it feel helpful, or did it feel like the interface was doing something without your input?*
- *Would you want a notification when the interface changes, or would you prefer it to just happen?*

Q 3

RESEARCHER
"How do you feel about the AI adjusting the interface layout without you explicitly clicking a settings button?"

→ PROBES
- *Is the level of control appropriate — or do you feel like you're losing agency?*

- *Would you want to be able to undo it easily?*

Q 4

RESEARCHER
"If you saw a message saying 'Interface adapted for focus — Accept?' — what would you do?"

→ PROBES
- *Would you accept without reading the details?*
- *What information would help you make that decision quickly?*
- *Would your answer change if you were in the middle of a stressful task?*

Q 5

RESEARCHER
"If you could type a plain-language instruction to change the interface — like 'make the text bigger' or 'hide the sidebar' — how does that compare to going into a Settings menu?"

→ PROBES
- *More intuitive, or less trustworthy?*
- *Would you actually use it, or would you still prefer the Settings menu?*
- *Does it feel more like talking to an assistant, or more like programming?*

→ TRANSITION
This connects to a broader idea I'd like to explore with you — the concept of an AI that works silently in the background all the time.

5 AI as Embedded Interface Assistant

BOTH PARTICIPANTS · ~5 MIN

Q 1

RESEARCHER
"How would you feel about an AI that runs silently in the background, observing how you interact — things like your mouse movements and reading speed — and uses that to adjust the interface?"

→ PROBES
- *Does the idea of being 'observed' feel uncomfortable?*
- *Would it matter if the data never left your device?*
- *Would you want to be able to see what it's collected?*

Q 2

RESEARCHER
"Would you want to be told every time the interface adapts, or would you prefer it to happen silently?"

→ PROBES
- *Would it depend on how big the change is?*
- *Would a subtle visual cue — like a small icon — be enough?*

Q 3

RESEARCHER
"Is there a point at which AI interface adaptation would feel too controlling or intrusive? Where would you draw that line?"

→ PROBES
- *Is changing the visual layout okay, but changing the content or order of your emails not okay?*
- *What if it hid emails it thought were unimportant — would that be too far?*
- *Would you want a 'manual override' always available?*

Q 4

RESEARCHER
"Thinking about your day-to-day email experience — which adaptations would add the most value for you? For example: font size, colour scheme, priority sorting, or email summarisation?"

→ PROBES
- *Can you rank them?*
- *Is there something not on that list that you'd want?*

Q 5

RESEARCHER
"If you were designing the perfect adaptive email assistant for yourself — with no technical constraints — what would it do?"

→ PROBES

- *No wrong answers here — go wild.*
- *Does it change based on your mood? Does it draft replies for you? Does it block distractions?*

→ TRANSITION

That's really insightful. We're almost done — just a couple of closing questions.

6 Closing

BOTH PARTICIPANTS · ~2 MIN

Q 1

RESEARCHER

"Is there anything you expected me to ask about, but I didn't?"

→ PROBES

- *Feel free — this is your chance to add anything we haven't covered.*

Q 2

RESEARCHER

"Is there anything else about the prototype or the concept of adaptive interfaces that you'd like to share?"

Closing statement

RESEARCHER

"Thank you so much. This has been incredibly helpful — your insights will directly shape how we develop this system further. I'll stop the recording now. Just to confirm — if I have any follow-up questions, is it okay to reach out by email?"

◆ *Stop recording. Thank participant again. Offer to share research findings when published.*

RESEARCHER NOTES TEMPLATE

Field sheet

Participant ID: ☐ P1-ADHD ☐ P2-Neurotypical

Date & Time: _____

Session Duration: _____

Prototype tested: ☐ BaseCase_test ☐ Dual-AI Version_test ☐ Both

Email Set: ☐ Set A (PhD Thesis) ☐ Set B (Coursework)

Key verbatim quotes (note timestamp if possible)

_____

_____

_____

Observed reactions during prototype demo

_____

_____

_____

Strongest pain points mentioned

_____

_____

Comfort level with AI [1=very uncomfortable … 5=very comfortable]:

_____

Most desired adaptation feature

_____

_____

_____

Surprises or unexpected responses

_____

_____

_____

Follow-up questions for next session

_____

_____

_____

Interview Script v1.0 · For internal research use only · DCAF Adaptive Email Client Study

C TaskSheet

USER STUDY — TASK ANSWER SHEET

Email Set A · PHD Thesis Track (Baseline email client)

Participant ID	Condition	Date	Session Order
_____	_____	_____	_____

INSTRUCTIONS TO PARTICIPANT: Read each task carefully, find the relevant email(s) in the interface, and write your answer in the space provided. A researcher will tell you when to start and stop each task. Work at your own pace — there are no trick questions. Your honesty is more valuable than a perfect score.

PART A — Tasks

Complete each task using the email interface provided. Write your answers below each task.

T1 **Information Retrieval** *Est. time: ~1 min*

Read the emails in your inbox. What is the EXACT deadline that Prof. Chen has given you to submit your Chapter 3 thesis draft? State both the date AND the time.

Task Start Time: ___ : ___ | *Task End Time: ___ : ___*

Your Answer:

T2 **Priority Ranking** *Est. time: ~2 min*

Rank the following three emails from HIGHEST to LOWEST priority. Write the ranking (1st / 2nd / 3rd) in the space below and explain your reasoning in 1–2 sentences per item.

- Email from the Research Ethics Committee
- Email from Dr. Sarah Mitchell (research collaboration)
- Email from the PhD Symposium Committee

Task Start Time: ___ : ___ | *Task End Time: ___ : ___*

Your Answer:

T3 **Email Composition** *Est. time: ~3 min*

Write a brief reply (2–4 sentences) to Dr. Sarah Mitchell's email. Your reply must: (1) confirm that you are available Thursday afternoon, and (2) ask her to confirm the meeting location or room.

Task Start Time: ___ : ___ | *Task End Time: ___ : ___*

Your Answer:

T4 **Cross-Email Integration** *Est. time: ~4 min*

Read ALL emails in the inbox. List EVERY action item you need to complete. For each item, state what needs to be done and by when (if a deadline is given). You may use bullet points.

Task Start Time: ___ : ___ | *Task End Time: ___ : ___*

Your Answer:

T5 **Decision Making** *Est. time: ~2 min*

Read the email from Alex Hartmann about the lab equipment booking. This email does NOT include an explicit deadline. Based on the content of the email, does this email require an immediate reply TODAY? Write your decision (Yes / No) and explain your reasoning in 2–3 sentences.

Task Start Time: ___ : ___ | *Task End Time: ___ : ___*

Your Answer:

Part B — NASA-TLX (Cognitive Load)

Instructions: For each dimension below, circle or write a number from 0 to 100 that best reflects your experience during the task session just completed. (0 = extremely low / not at all; 100 = extremely high / very much so). Note: for 'Performance', a score of 0 means very satisfied and 100 means very dissatisfied.

Dimension	Description	Your Rating (0 – 100)
Mental Demand	How much thinking, judging, calculating, or remembering did this task require?	_____
Temporal Demand	How rushed or hurried was the pace of the task? Did you feel time pressure?	_____
Performance	How satisfied are you with your performance? (0 = very satisfied, 100 = very dissatisfied)	_____
Effort	How hard did you have to work (mentally and physically) to achieve your level of performance?	_____
Frustration	How irritated, stressed, annoyed, or frustrated did you feel during the task?	_____

Total NASA-TLX Score (researcher calculates): _____

Formula: Average the 5 ratings, then reverse Performance (100 - score) before averaging. Final score range: 0–100. Higher = greater cognitive load.

Part C — System Usability Scale (SUS)

Instructions: For each statement below, circle or write a number from 1 to 5 where:
1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

#	Statement	Rating (1–5)
1	I think that I would like to use this system frequently.	_____
2	I found the system unnecessarily complex.	_____
3	I thought the system was easy to use.	_____

4	I think that I would need the support of a technical person to be able to use this system.	_____
5	I found the various functions in this system were well integrated.	_____
6	I thought there was too much inconsistency in this system.	_____
7	I would imagine that most people would learn to use this system very quickly.	_____
8	I found the system very cumbersome to use.	_____
9	I felt very confident using the system.	_____
10	I needed to learn a lot of things before I could get going with this system.	_____

SUS Score (researcher calculates): _____

Formula: Odd items (1,3,5,7,9): score − 1. Even items (2,4,6,8,10): 5 − score. Sum all 10 adjusted scores × 2.5 = SUS score (0–100). Score ≥ 68 = acceptable usability.

Part D — Dual-AI Adaptive Interface Questions

FILL IN THIS SECTION ONLY IF YOU TESTED THE DUAL-AI ADAPTIVE VERSION.
Rate each statement 1–5 (1 = Strongly Disagree, 5 = Strongly Agree), then answer the open questions in the space below each one.

Q1 The interface adaptations that occurred felt appropriate to what I was doing at the time.

Rating (1–5):

Your Answer:

Why? (open response):

Your Answer:

Q2 Even when the interface adapted automatically, I still felt in control of my email experience.

Rating (1–5):

Your Answer:

Why? (open response):

Your Answer:

Q3 Being able to describe my preferences in natural language felt more natural than going into a Settings menu.

Rating (1–5):

Your Answer:

Did you actually use the natural language input? If yes, what did you say? If no, why not?

Your Answer:

Q4 I would prefer using this adaptive interface over a standard email client for my daily work.

Rating (1–5):

Your Answer:

Why? What would you change?

Your Answer:

Open-Ended Follow-Up Questions (Researcher asks aloud)

The researcher will ask these questions verbally. Write key points from the participant's response.

OQ1 *When the interface changed automatically, what was your first reaction? Did you notice what specifically changed?*

Your Answer:

OQ2 *Was there any adaptation that felt unnecessary or annoying? Which one, and why?*

Your Answer:

OQ3 *If you used this system daily, what is the first thing you would want the AI to learn about you?*

Your Answer:

OQ4 *The notification auto-dismisses after 20 seconds. Did that feel right — too fast, too slow? What would you prefer?*

Your Answer:

OQ5 *Would you recommend this system to a friend or colleague? Why or why not?*

Your Answer:

USER STUDY — TASK ANSWER SHEET

Email Set B · Coursework & Research Project (Dual-AI adaptive client)

Participant ID	Condition	Date	Session Order
_____	_____	_____	_____

INSTRUCTIONS TO PARTICIPANT: Read each task carefully, find the relevant email(s) in the interface, and write your answer in the space provided. A researcher will tell you when to start and stop each task. Work at your own pace — there are no trick questions. Your honesty is more valuable than a perfect score.

PART A — Tasks

Complete each task using the email interface provided. Write your answers below each task.

T1	**Information Retrieval**	*Est. time: ~1 min*

Read the emails in your inbox. According to Professor Williams' email:

- (a) What percentage of the final course grade does the group project presentation account for?
- (b) What is the exact submission deadline for the presentation slides (date and time)?

Task Start Time: ___ : ___	*Task End Time: ___ : ___*

Your Answer:

T2	Priority Ranking	Est. time: ~2 min

Rank the following three emails from HIGHEST to LOWEST priority. Write the ranking (1st / 2nd / 3rd) below and explain your reasoning in 1–2 sentences per item.

- Email from Prof. James Lee (Assignment 2 extension)
- Email from University Library (overdue book)
- Email from Graduate Student Union (Research Showcase applications)

Task Start Time: ___ : ___	Task End Time: ___ : ___

Your Answer:

T3	Email Composition	Est. time: ~3 min

Write a brief reply (2–4 sentences) to Priya Sharma's email. Your reply must: (1) confirm you are available Tuesday and Wednesday afternoon, but NOT Thursday, and (2) suggest a specific meeting time from those available slots.

Task Start Time: ___ : ___	Task End Time: ___ : ___

Your Answer:

T4	Cross-Email Integration	Est. time: ~4 min

Read ALL emails in the inbox. List EVERY action item you need to complete. For each item, state what needs to be done and by when (if a deadline is given). You may use bullet points.

Task Start Time: ___ : ___	Task End Time: ___ : ___

Your Answer:

T5	Decision Making	Est. time: ~2 min

Read the email from the Department Research Coordinator about the ethics guidelines. This email does NOT state a clear deadline or required action date. Based on the content of the email, does this email require an immediate reply TODAY? Write your decision (Yes / No) and explain your reasoning in 2–3 sentences.

Task Start Time: ___ : ___	Task End Time: ___ : ___

Your Answer:

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Rating (1–5):

Your Answer:

Why? (open response):
Your Answer:

Q2 Even when the interface adapted automatically, I still felt in control of my email experience.
Rating (1–5):
Your Answer:

Why? (open response):
Your Answer:

Q3 Being able to describe my preferences in natural language felt more natural than going into a Settings menu.
Rating (1–5):
Your Answer:

Did you actually use the natural language input? If yes, what did you say? If no, why not?
Your Answer:

Q4 I would prefer using this adaptive interface over a standard email client for my daily work.
Rating (1–5):
Your Answer:

Why? What would you change?
Your Answer:

Open-Ended Follow-Up Questions (Researcher asks aloud)

The researcher will ask these questions verbally. Write key points from the participant's response.

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Your Answer:

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Your Answer:

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Your Answer:

OQ4 *The notification auto-dismisses after 20 seconds. Did that feel right — too fast, too slow? What would you prefer?*
Your Answer:

OQ5 *Would you recommend this system to a friend or colleague? Why or why not?*
Your Answer: